



US012399152B2

(12) **United States Patent**
Oxley et al.

(10) **Patent No.:** **US 12,399,152 B2**

(45) **Date of Patent:** **Aug. 26, 2025**

(54) **APPARATUS AND METHODS FOR PREPARATION AND INTRODUCTION OF TRACE SAMPLES INTO AN IONIZING DETECTION SYSTEM**

(71) Applicant: **University of Rhode Island Board of Trustees**, Kingston, RI (US)

(72) Inventors: **Jimmie Oxley**, Narragansett, RI (US);
James L. Smith, Narragansett, RI (US);
Alexander Yevdokimov, Norwich, CT (US)

(73) Assignee: **University of Rhode Island Board of Trustees**, Kingston, RI (US)

(*) Notice: Subject to any disclaimer, the term of this patent is extended or adjusted under 35 U.S.C. 154(b) by 757 days.

(21) Appl. No.: **17/715,287**

(22) Filed: **Apr. 7, 2022**

(65) **Prior Publication Data**

US 2022/0229017 A1 Jul. 21, 2022

Related U.S. Application Data

(63) Continuation-in-part of application No. 16/804,787, filed on Feb. 28, 2020, now Pat. No. 11,300,542.

(60) Provisional application No. 62/816,253, filed on Mar. 11, 2019.

(51) **Int. Cl.**

G01N 27/00 (2006.01)

G01N 27/622 (2021.01)

H01J 49/00 (2006.01)

H01J 49/26 (2006.01)

(52) **U.S. Cl.**

CPC **G01N 27/622** (2013.01); **H01J 49/0031** (2013.01); **H01J 49/26** (2013.01)

(58) **Field of Classification Search**

CPC G01N 27/622; G01N 2001/028; H01J 49/0409; H01J 49/0031; H01J 49/025; H01J 49/16

USPC 250/281, 282, 288

See application file for complete search history.

(56) **References Cited**

U.S. PATENT DOCUMENTS

8,704,167 B2 4/2014 Cooks et al.

8,710,437 B2 4/2014 Cooks et al.

8,859,986 B2 10/2014 Cooks et al.

(Continued)

FOREIGN PATENT DOCUMENTS

CN 102636553 B 5/2014

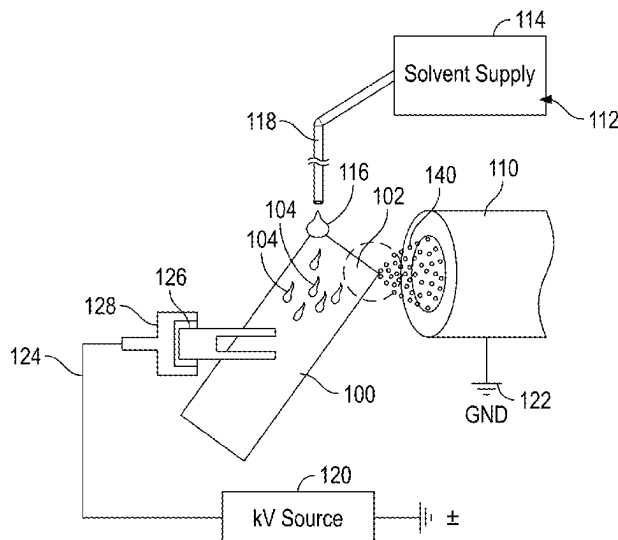
Primary Examiner — Michael Maskell

(74) *Attorney, Agent, or Firm* — Hinckley Allen & Snyder; Stephen Holmes

(57) **ABSTRACT**

The present disclosure describes an improved sample preparation technique, swab, swab holder and solvent for the removal of trace chemical materials from swabs and delivery of the trace materials into a mass spectrometer (MS) or an ion mobility spectrometer (IMS). The devices and techniques aid the delivery of the analyte into the instrument inlet and concurrently serves as an ionization source. The technique includes the steps of positioning or shaping the swab so that a pointed analyte release area is directed toward the inlet of the MS or IMS, adding a solvent to the swab and charging the swab to create an ionized analyte spray (Taylor Cone spray) into the inlet of the detection instrument. Solvent may include a chemical for depressing the ionization of residual solvent or enhancing the ionization of the desired analyte and this chemical may serve as an internal standard or benchmark.

22 Claims, 32 Drawing Sheets



(56)

References Cited

U.S. PATENT DOCUMENTS

8,895,918	B2 *	11/2014	Cooks	G01N 30/91 250/281
8,932,875	B2 *	1/2015	Cooks	H01J 49/0031 436/60
8,937,288	B1	1/2015	Cooks et al.	
9,035,239	B1	5/2015	Cooks et al.	
9,116,154	B2	8/2015	Ouyang et al.	
9,165,752	B2	10/2015	Cooks et al.	
10,088,461	B2	10/2018	Cooks et al.	
2013/0344610	A1 *	12/2013	Cooks	H01J 49/0431 436/141
2014/0183351	A1 *	7/2014	Cooks	H01J 49/167 250/288

* cited by examiner

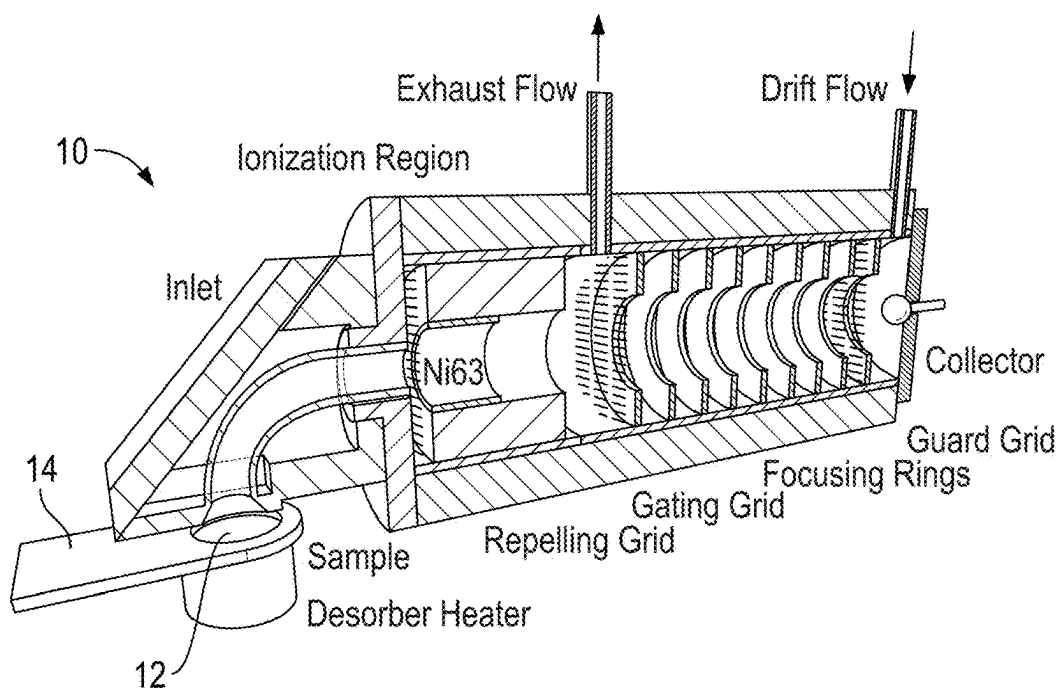


FIG. 1
(PRIOR ART)

Mass Spectrometer

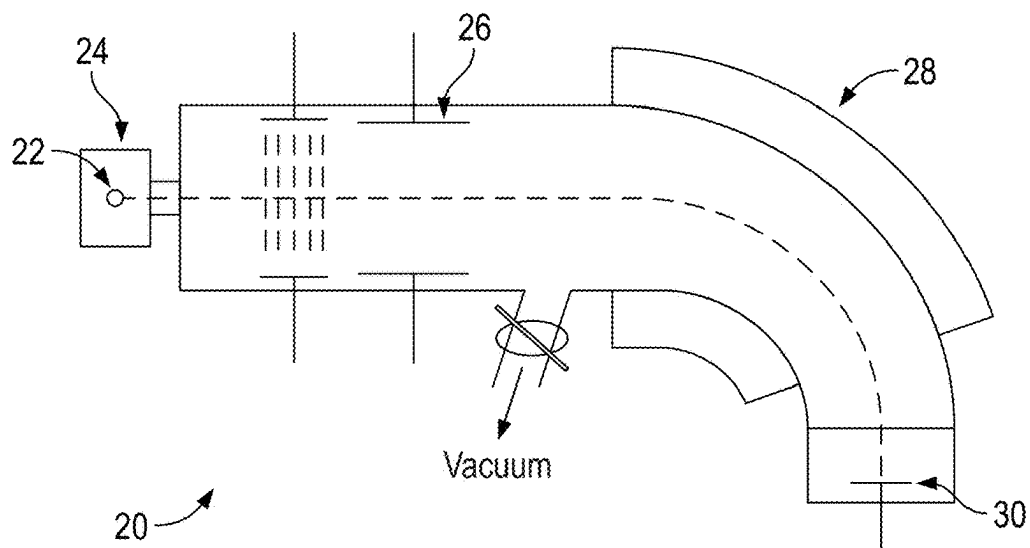


FIG. 2
(PRIOR ART)

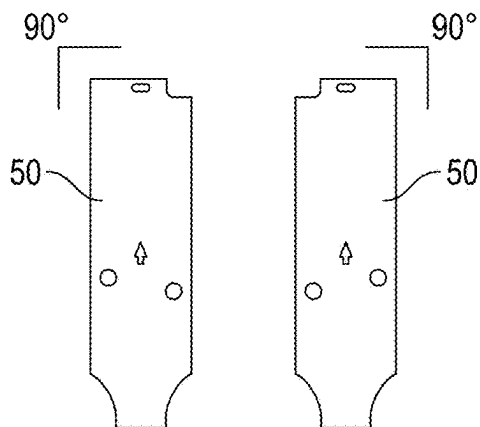


FIG. 3A
(PRIOR ART)

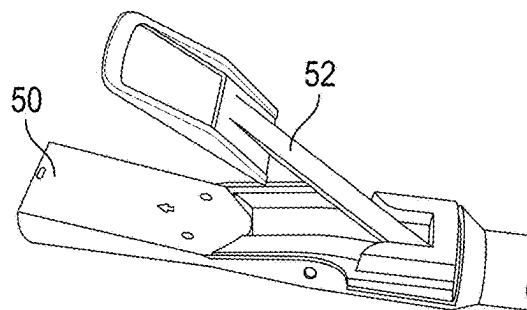


FIG. 3B
(PRIOR ART)

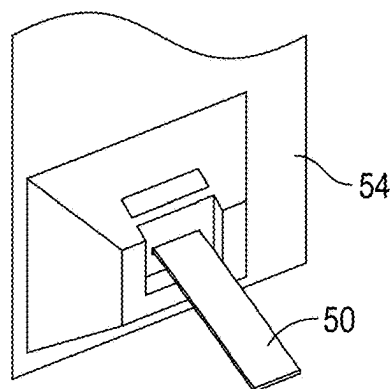


FIG. 3C
(PRIOR ART)

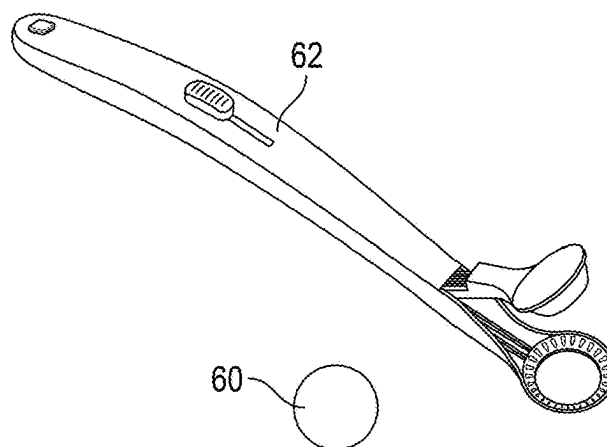


FIG. 4
(PRIOR ART)

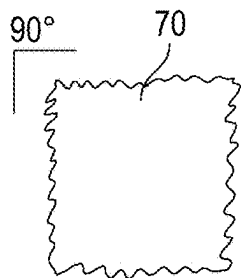


FIG. 5A
(PRIOR ART)

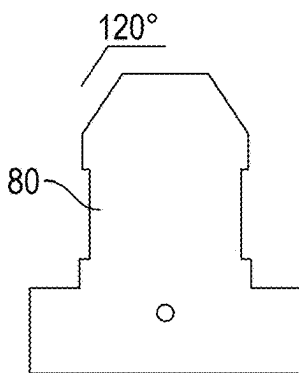


FIG. 5B
(PRIOR ART)

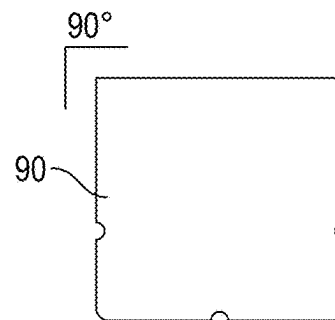


FIG. 5C
(PRIOR ART)

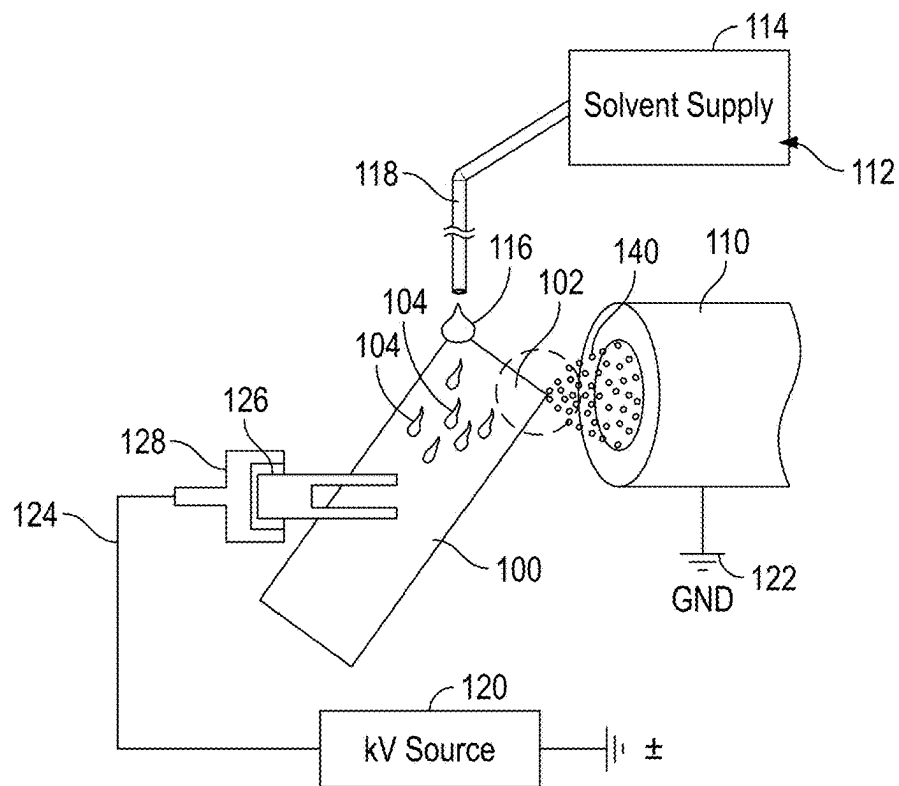


FIG. 6

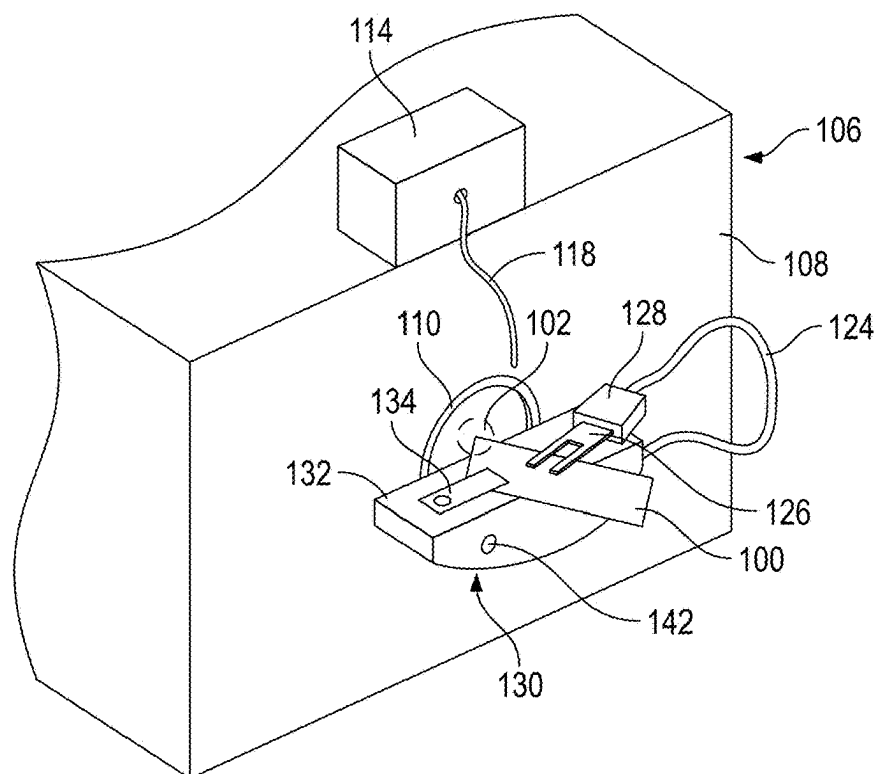


FIG. 7

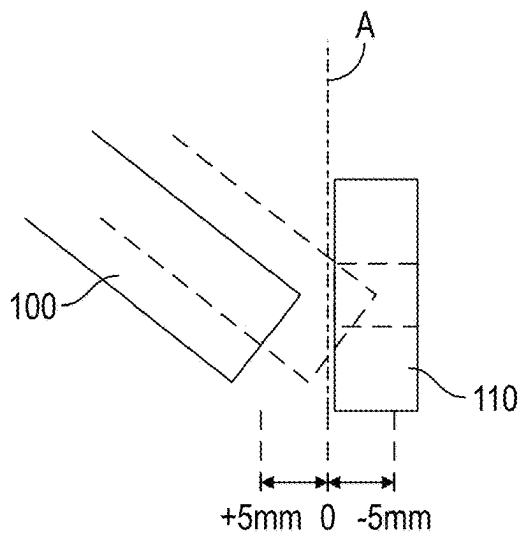


FIG. 8A

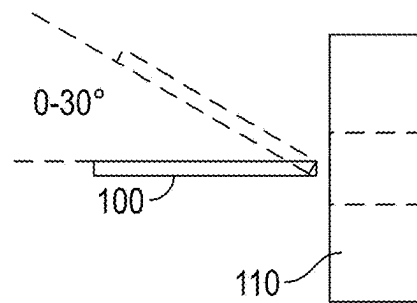


FIG. 8B

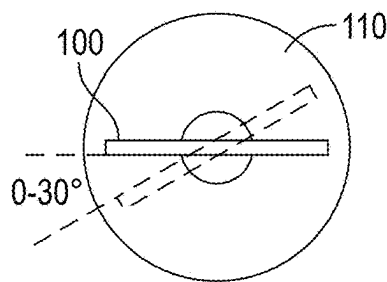


FIG. 8C

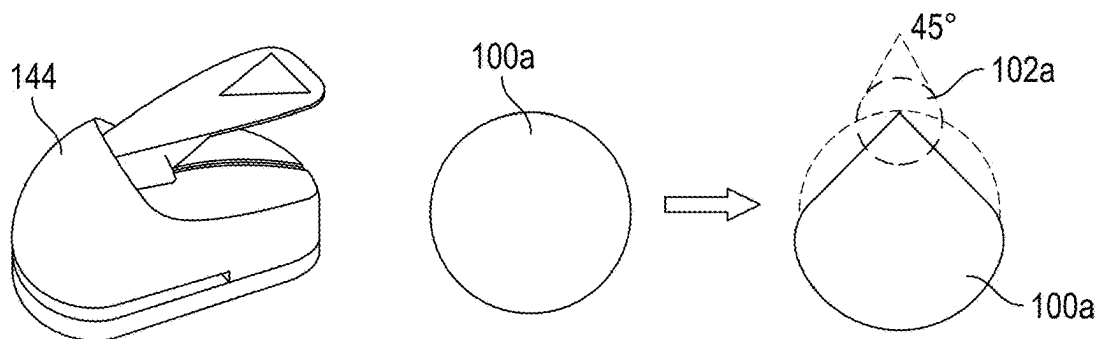


FIG. 9

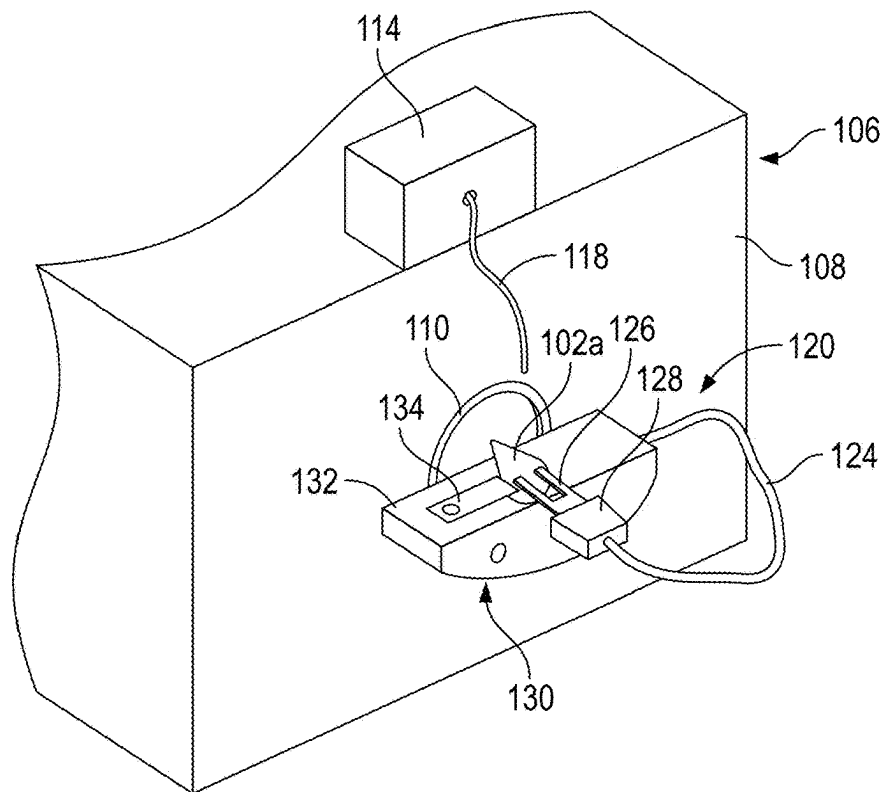


FIG. 10

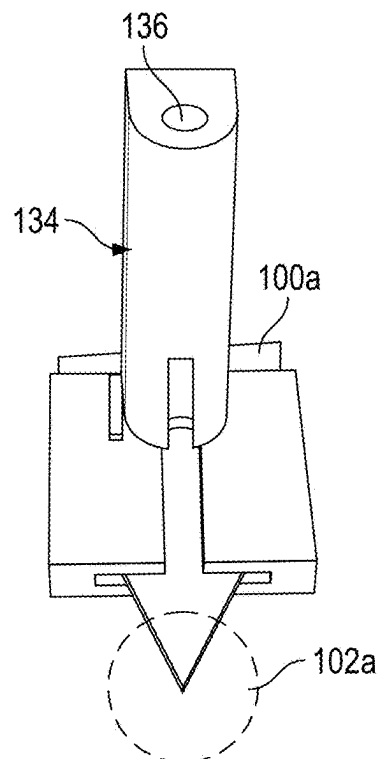


FIG. 11

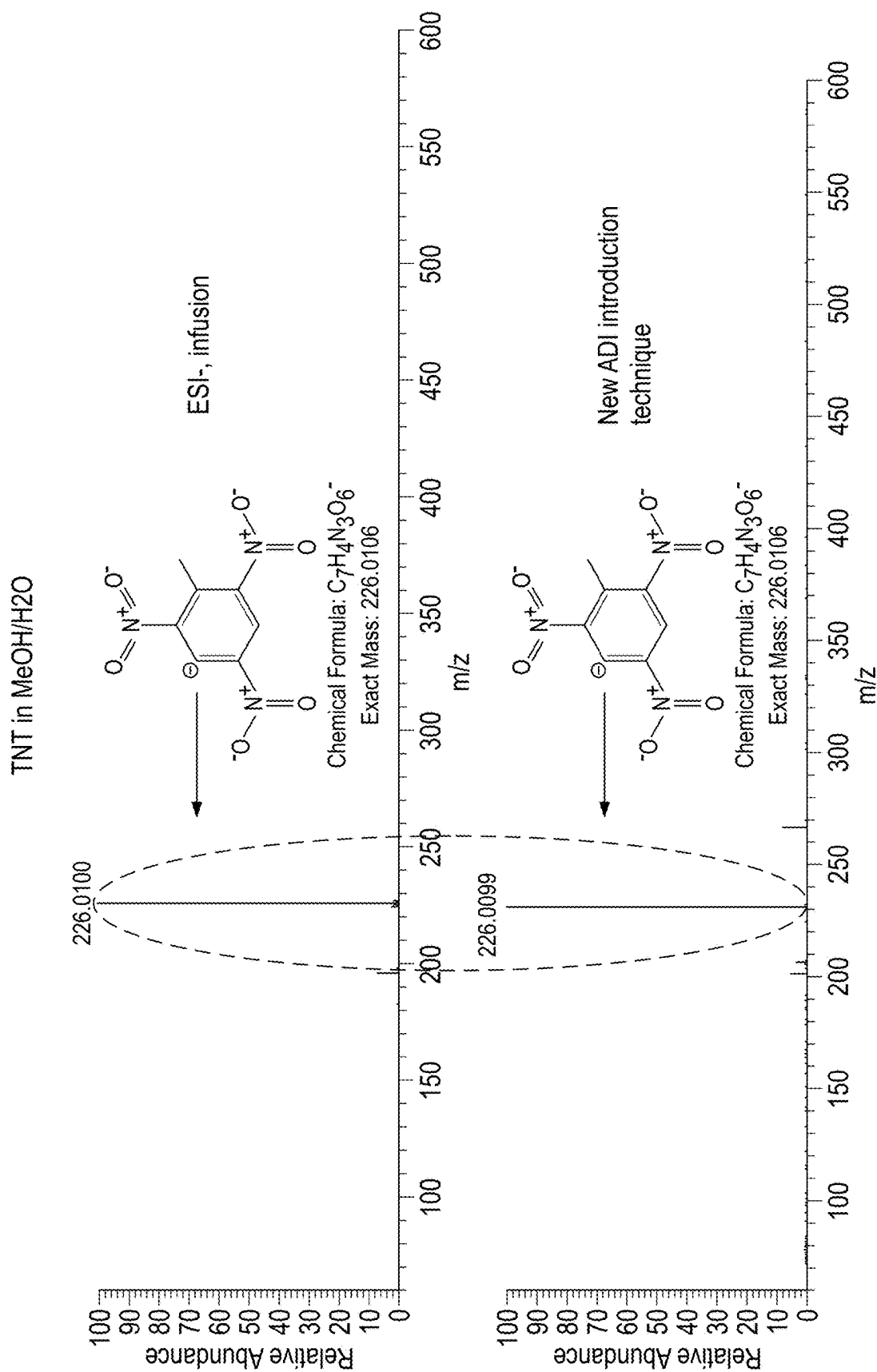


FIG. 12A

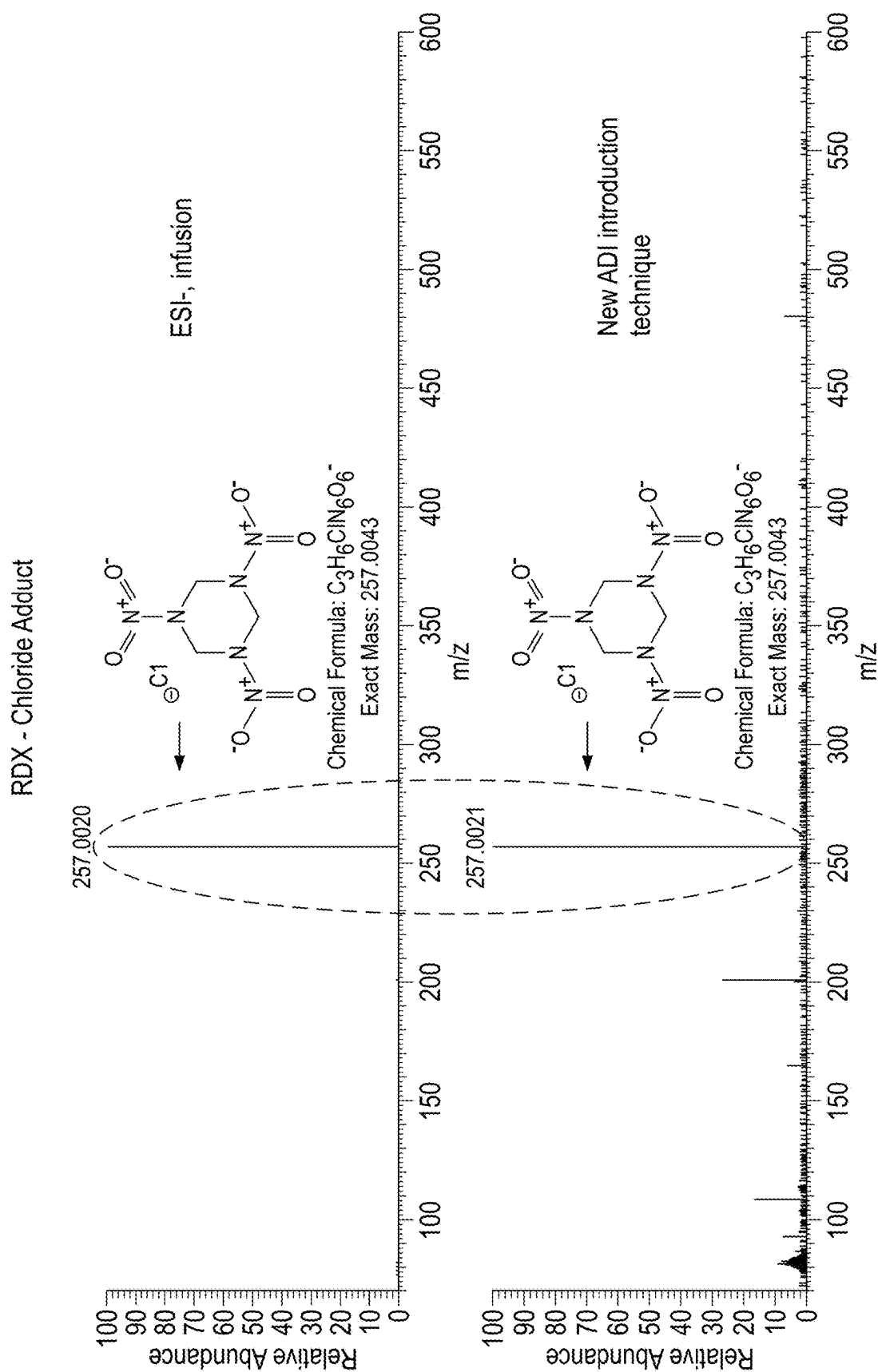


FIG. 12B

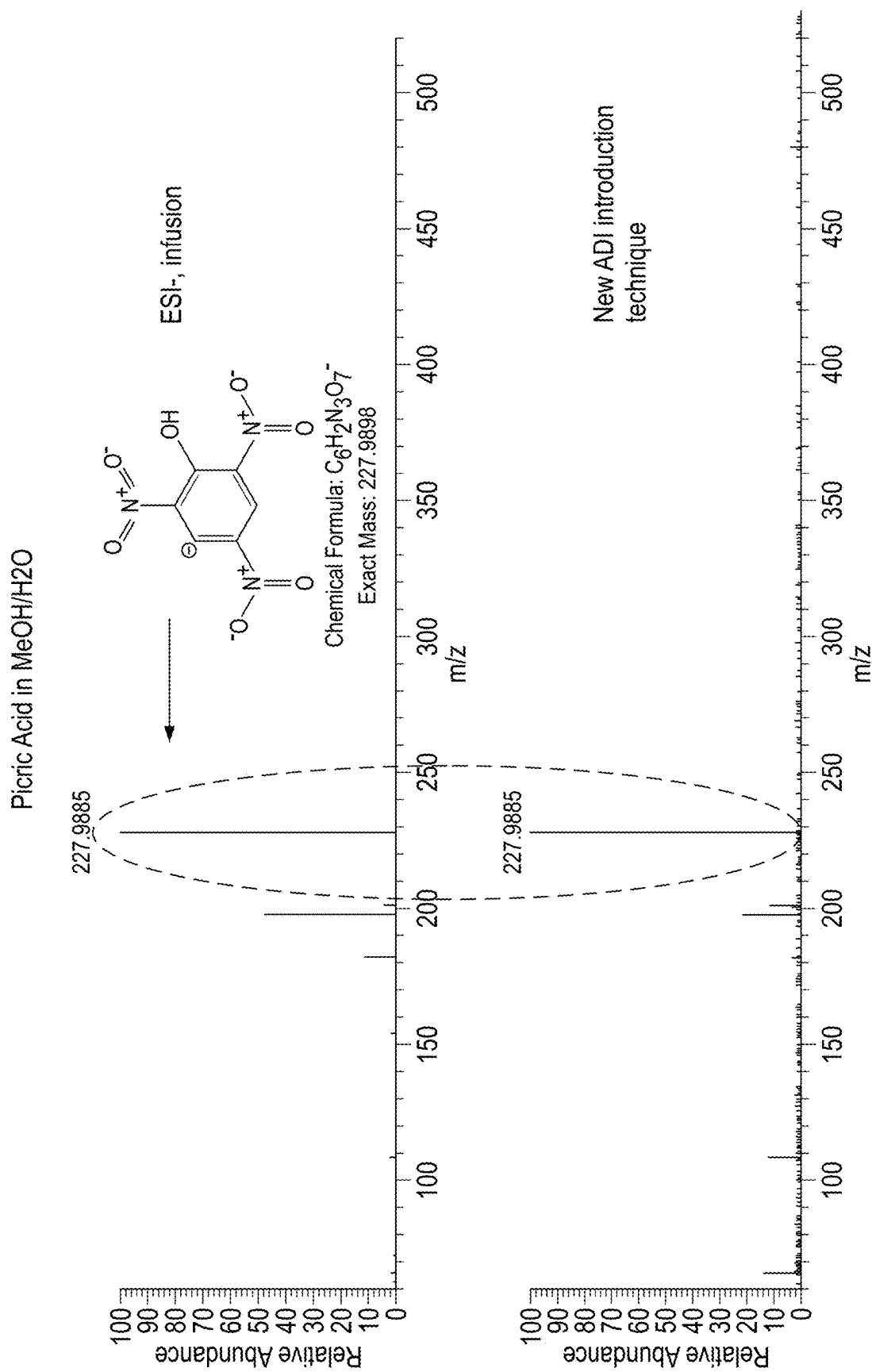


FIG. 12C

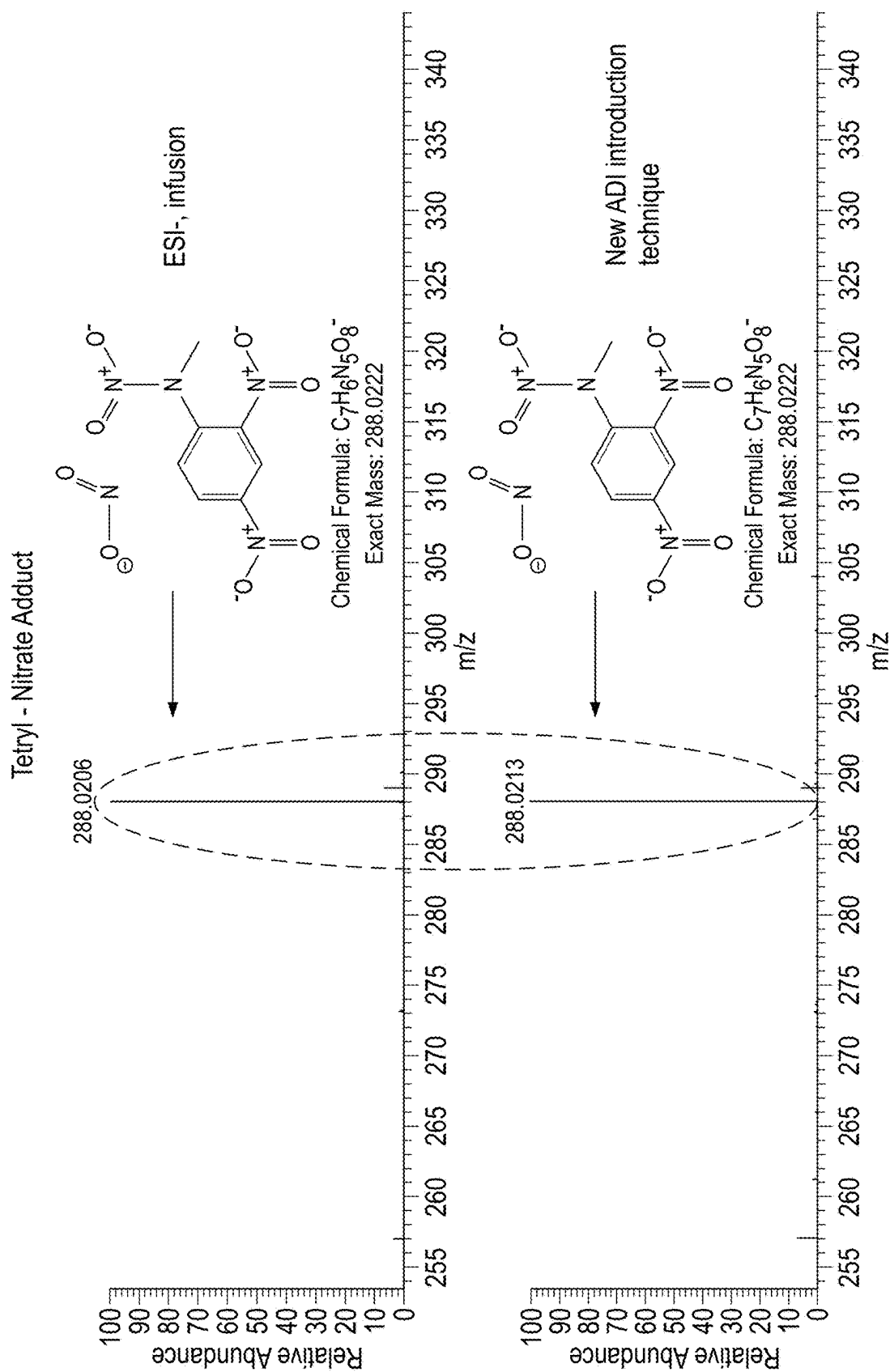


FIG. 12D

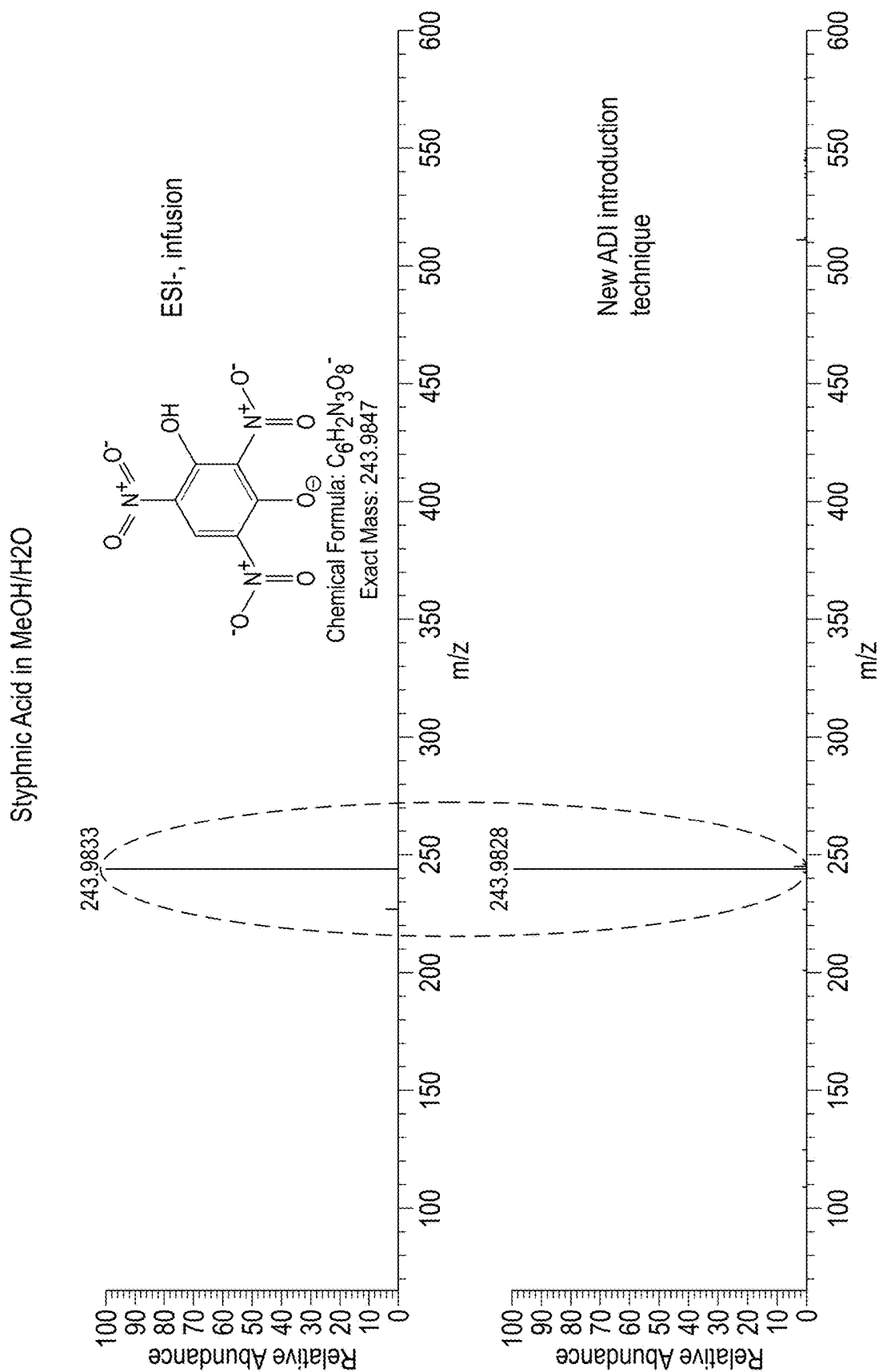


FIG. 13A

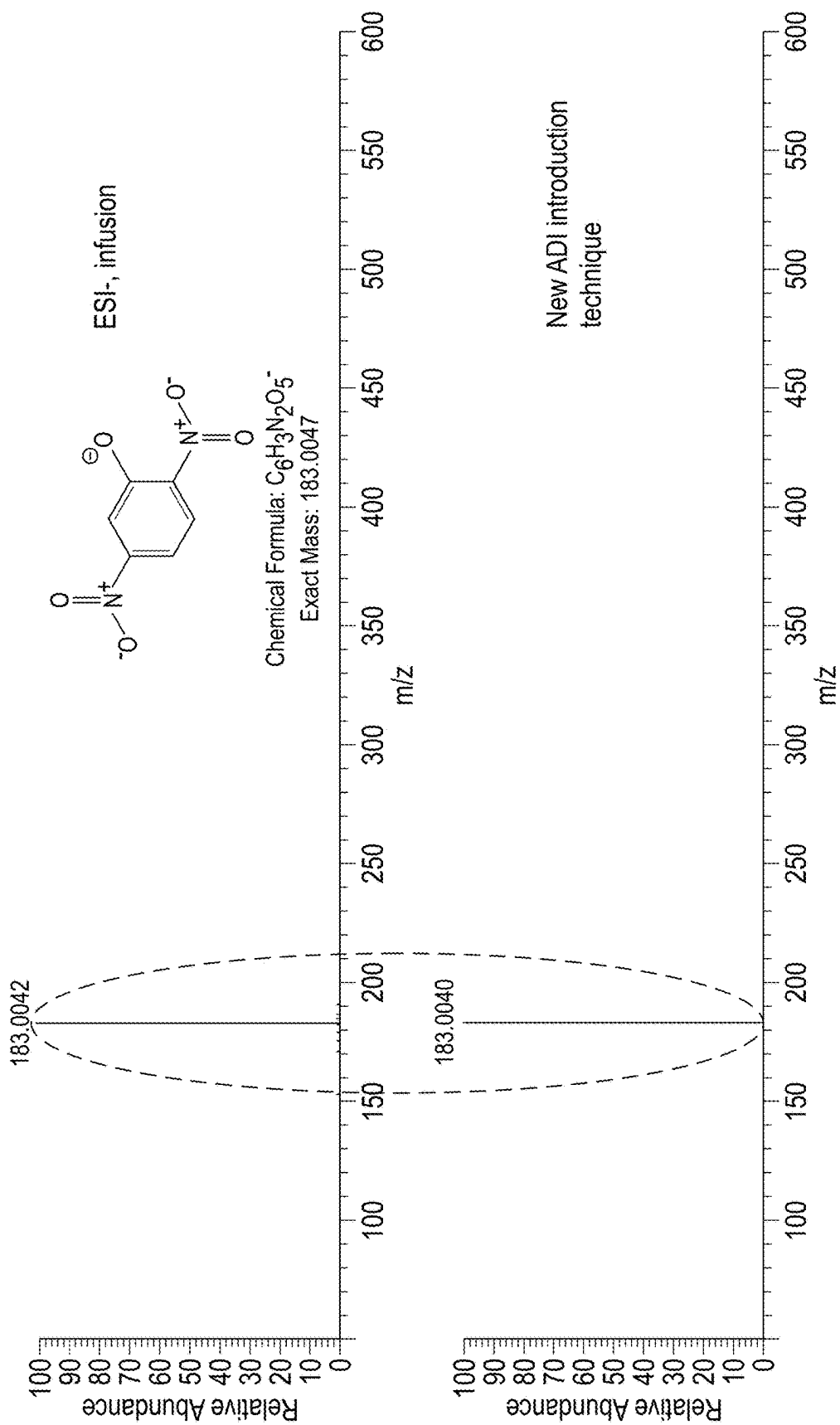
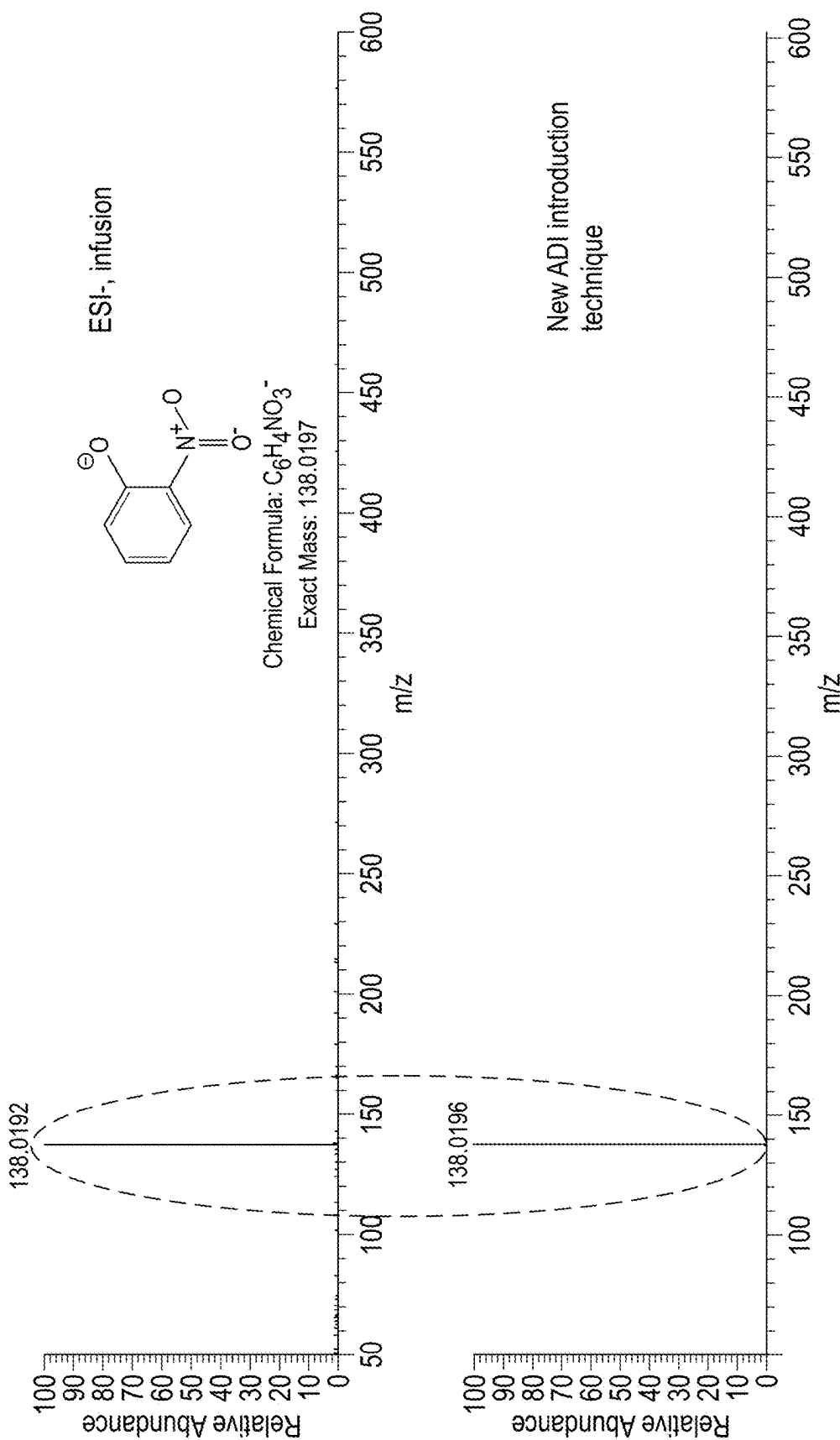
2,5 - Dinitrophenol in MeOH/H₂O

FIG. 13B

2 - Nitrophenol in MeOH/H₂O

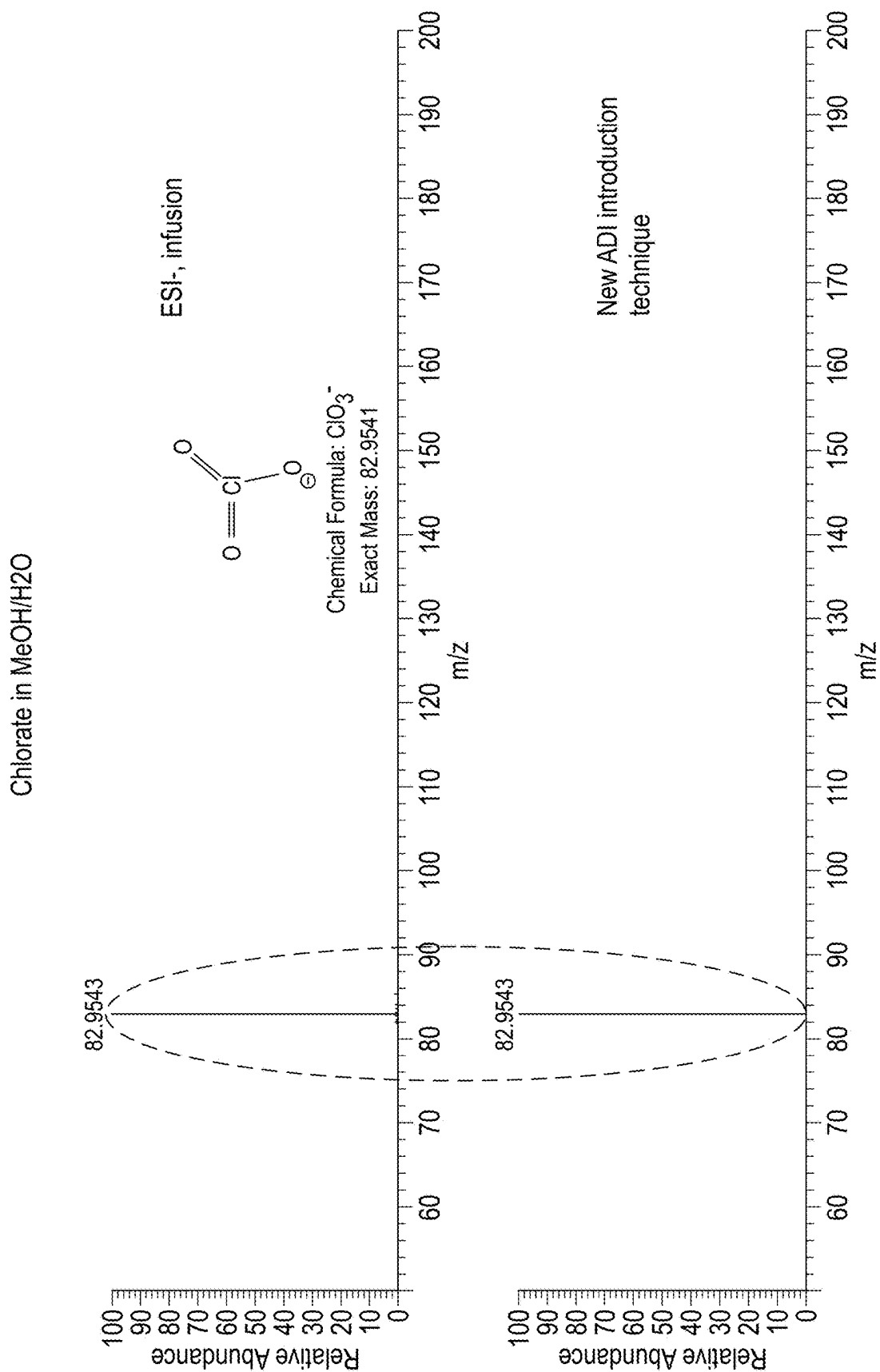


FIG. 13D

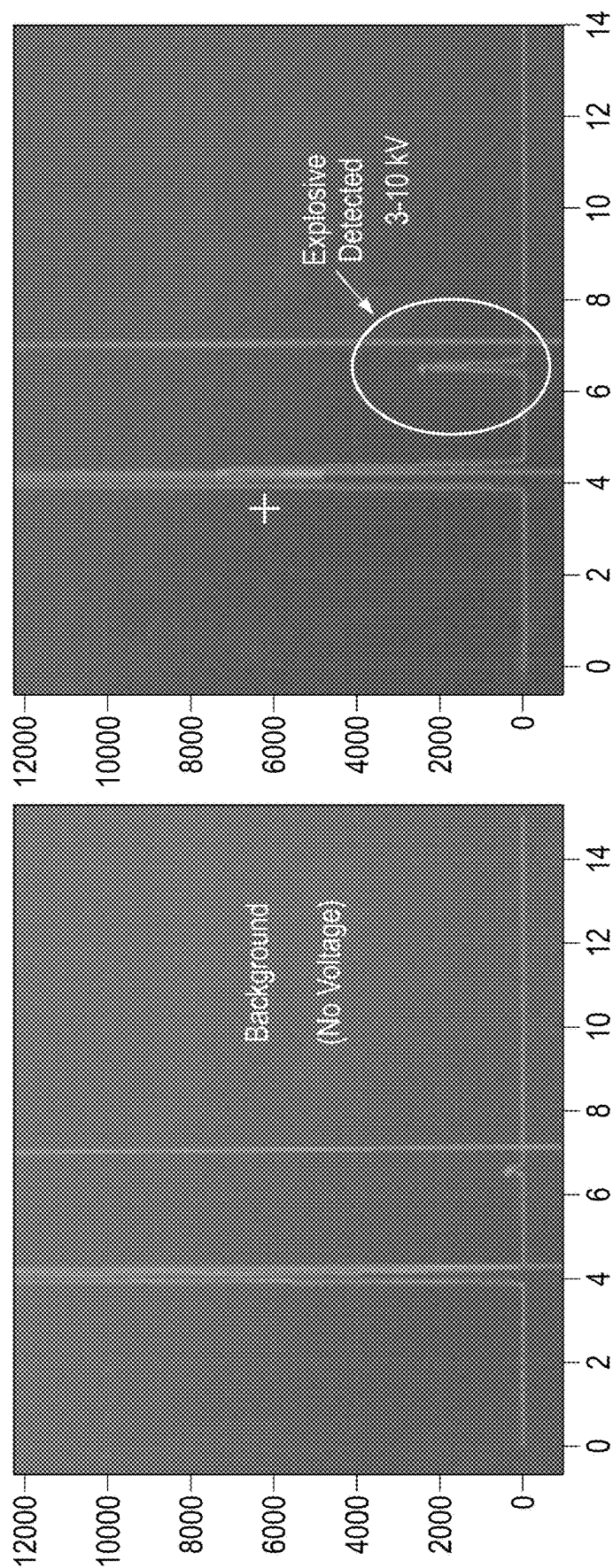


FIG. 14

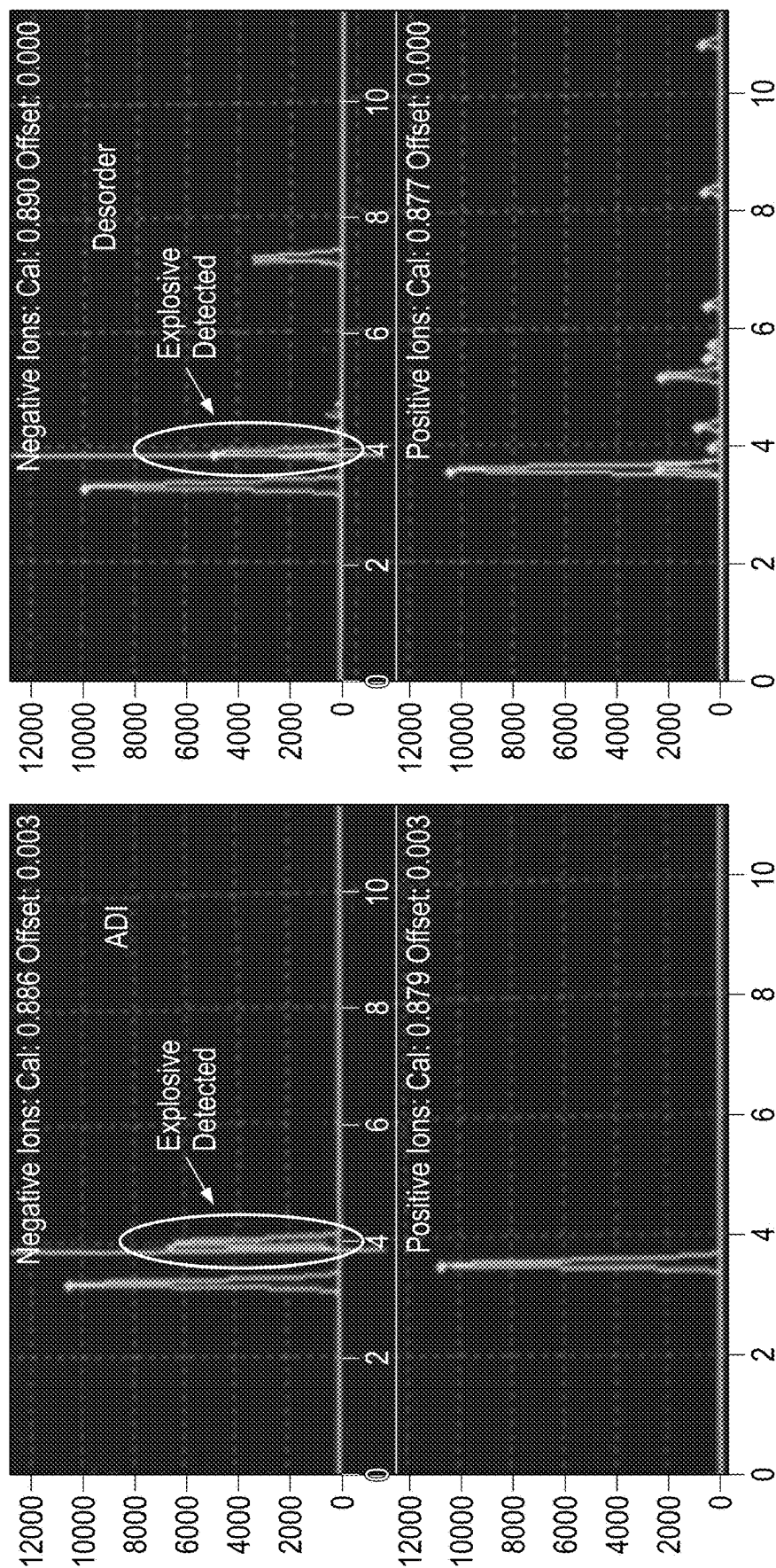


FIG. 15

Sample: Ammonium Nitrate
Solvent: Ethanol (100%)

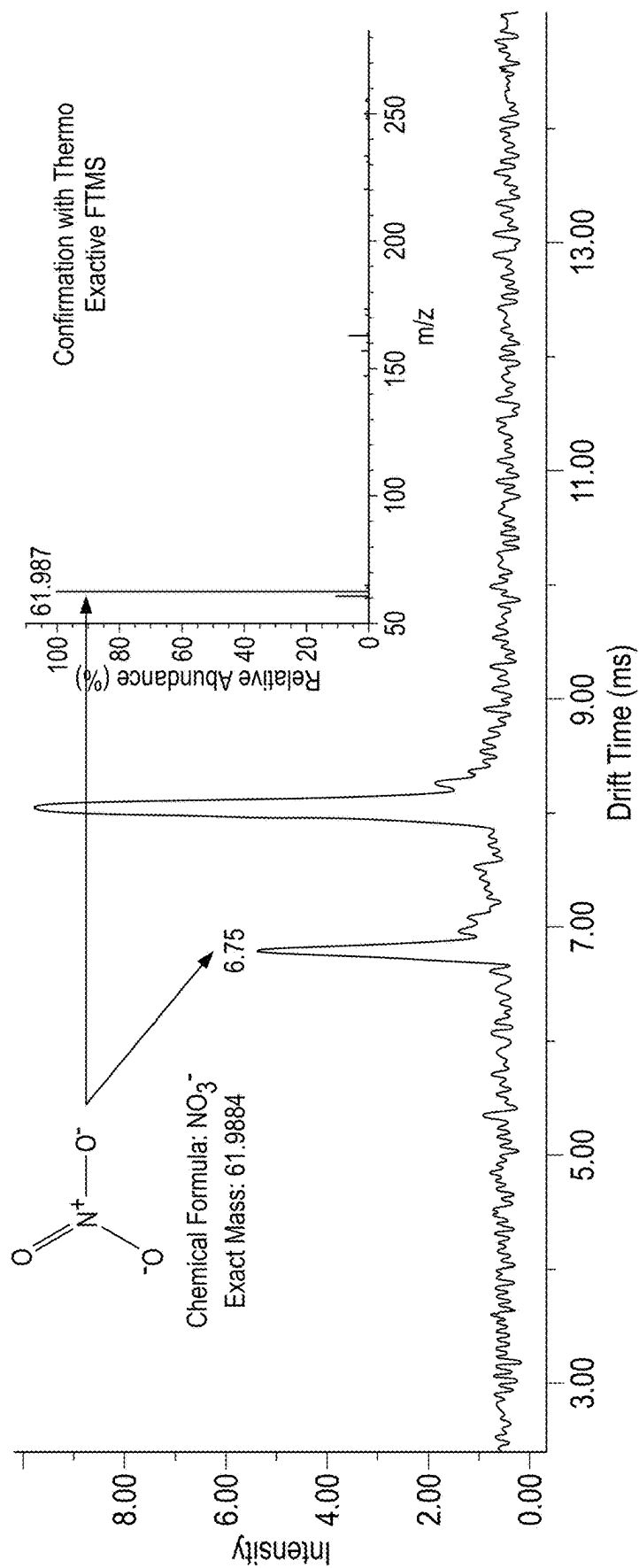


FIG. 16A

Sample: Sodium Chlorate
Solvent: Ethanol (100%)

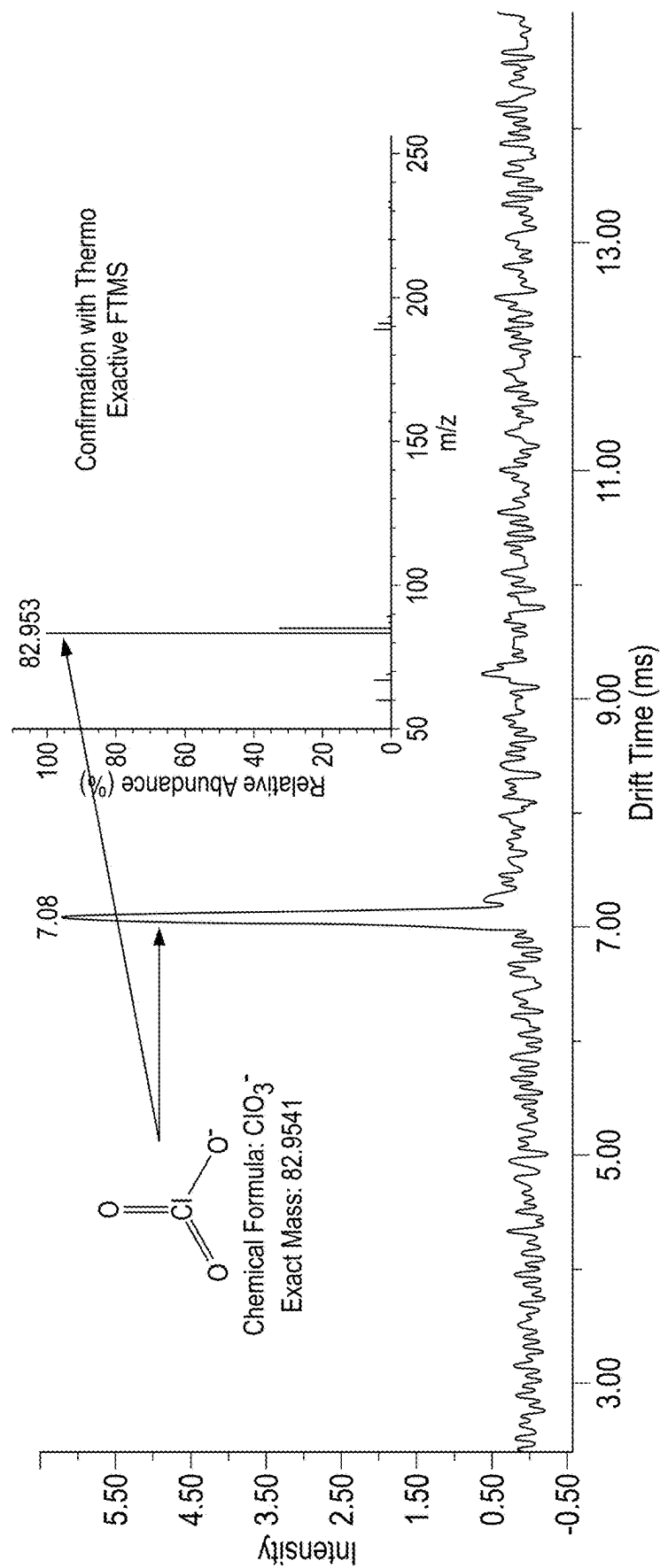


FIG. 16B

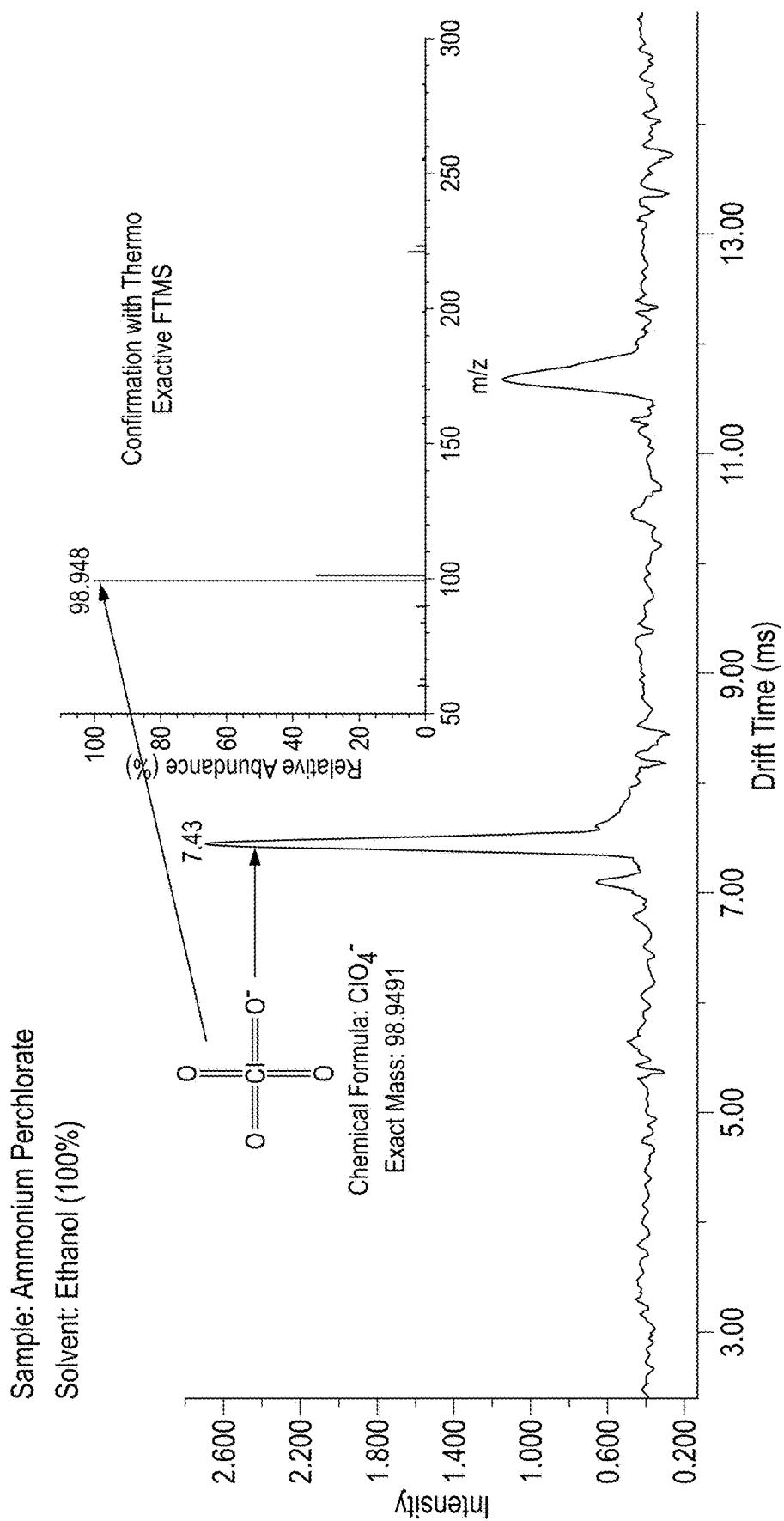


FIG. 16C

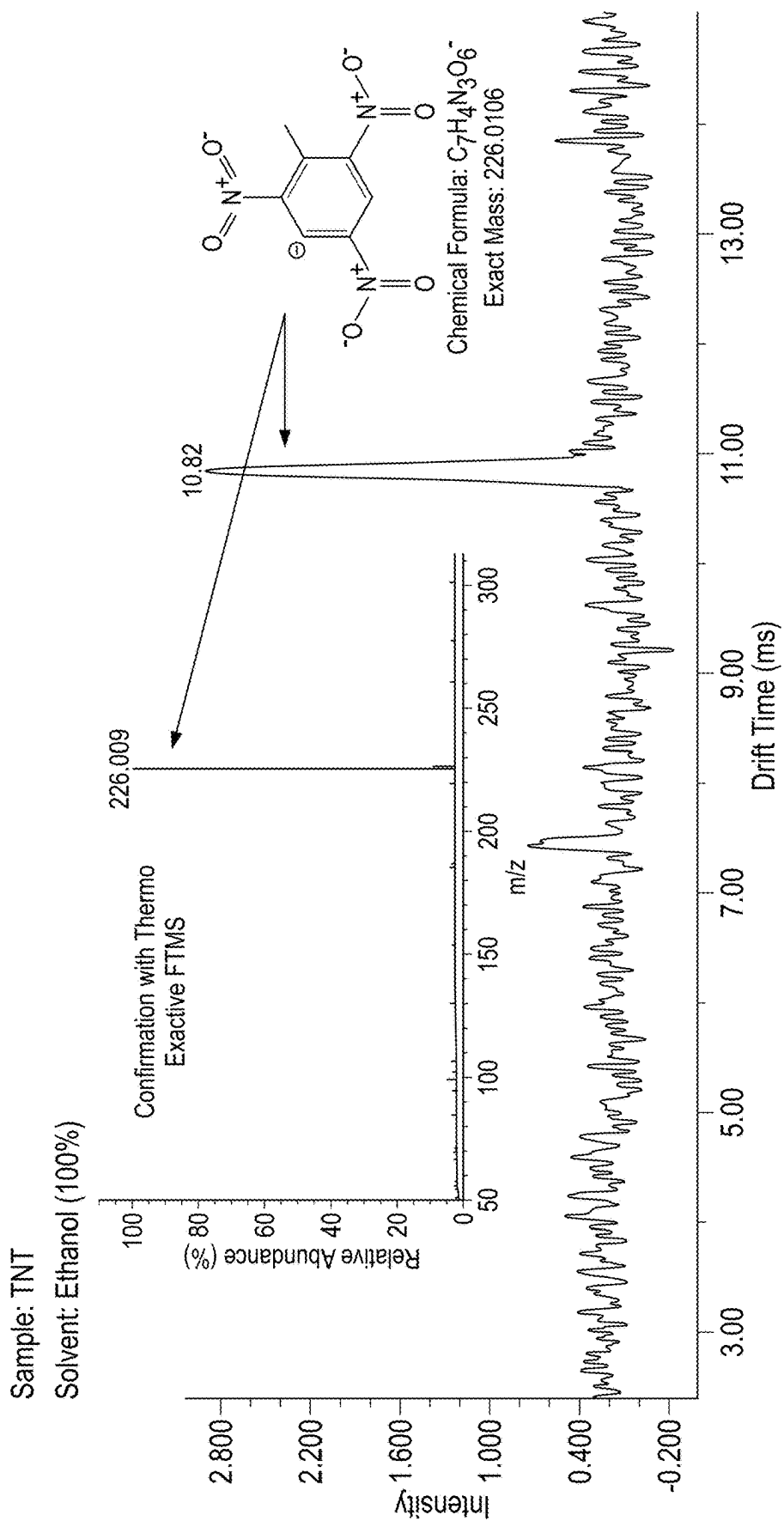


FIG. 16D

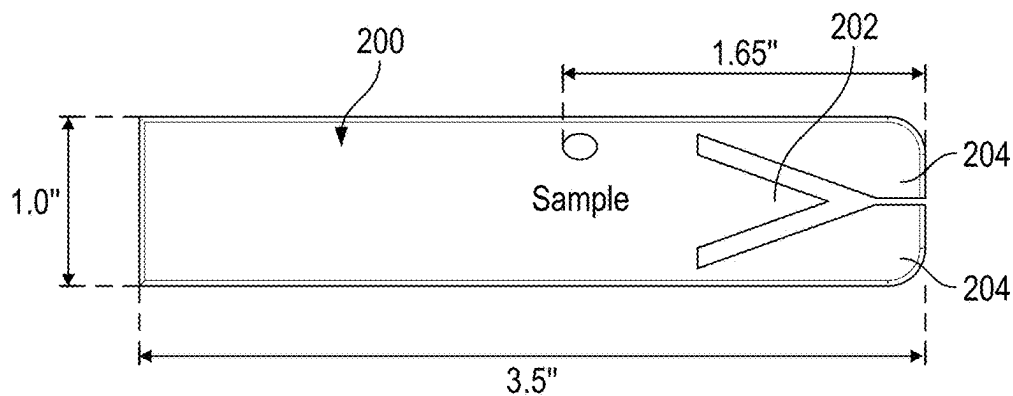


FIG. 17

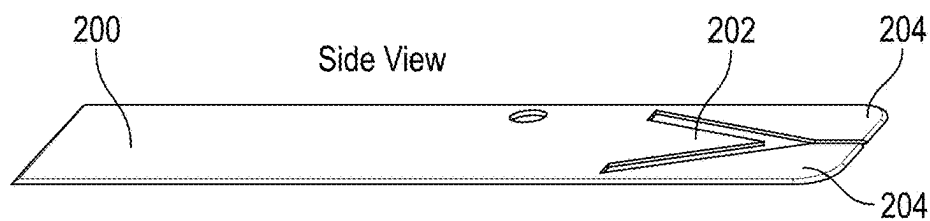


FIG. 18

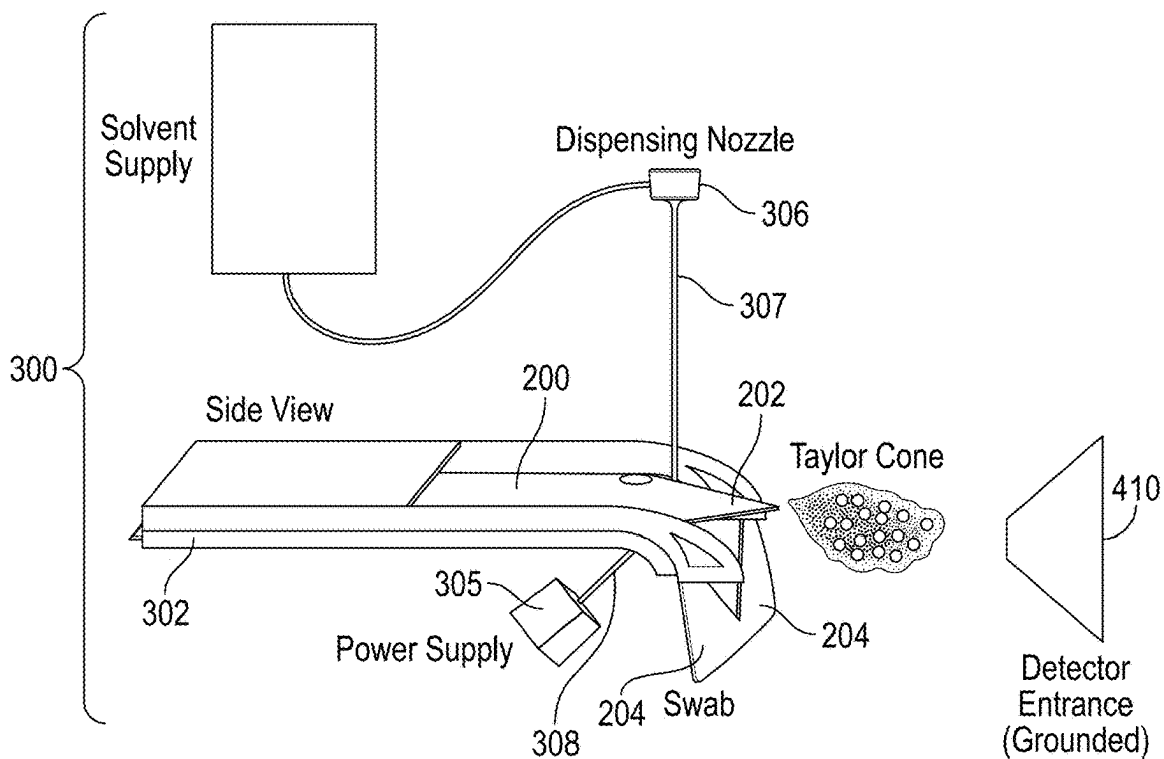


FIG. 19

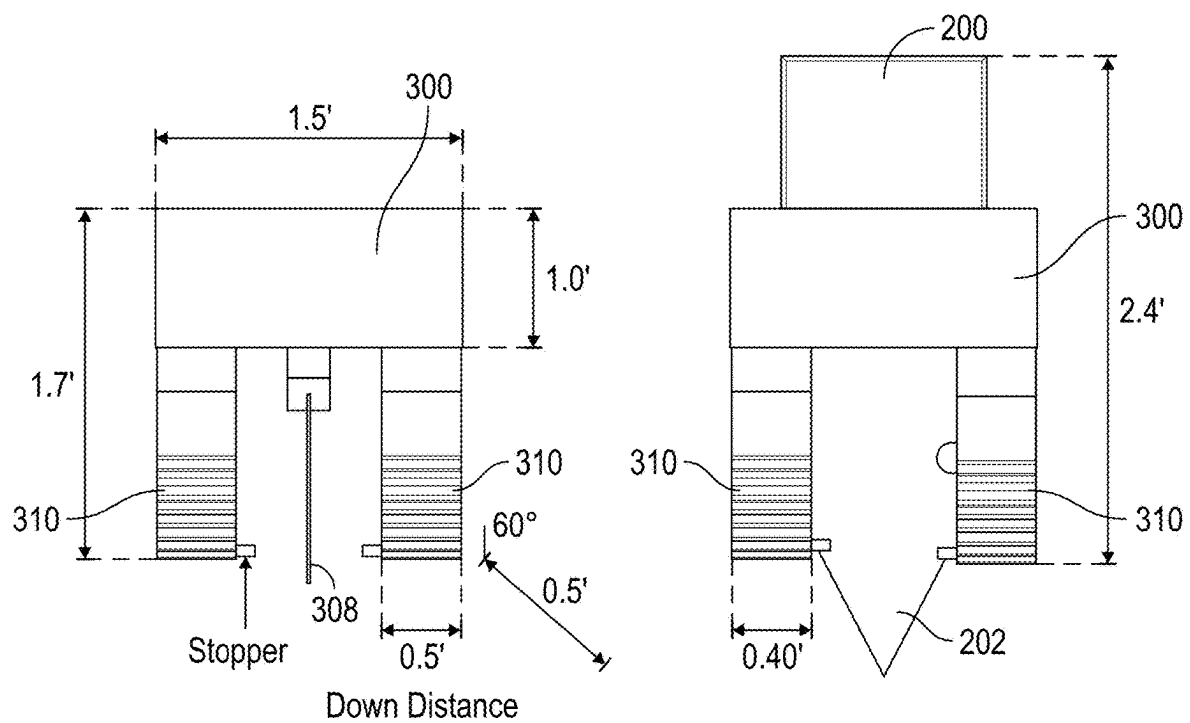


FIG. 20A

FIG. 20B

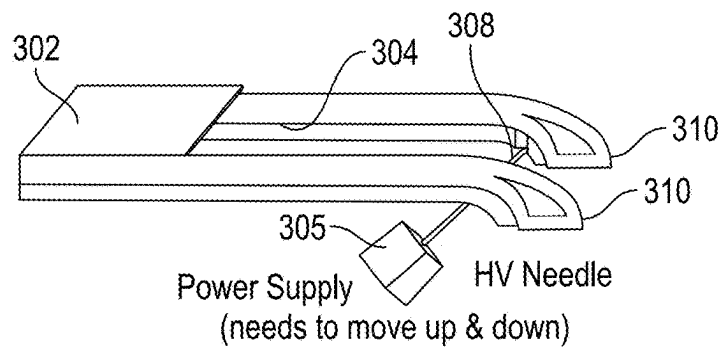


FIG. 20C

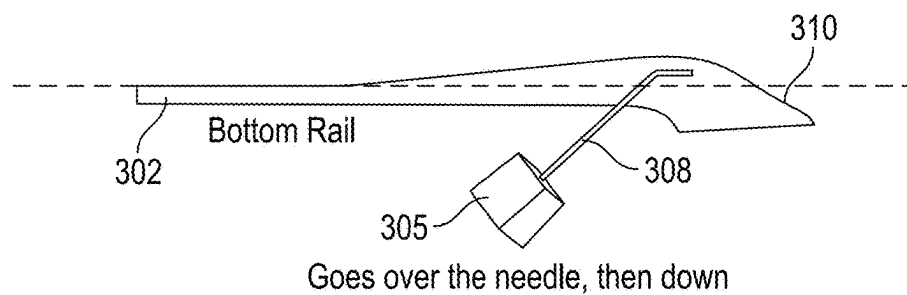


FIG. 20D

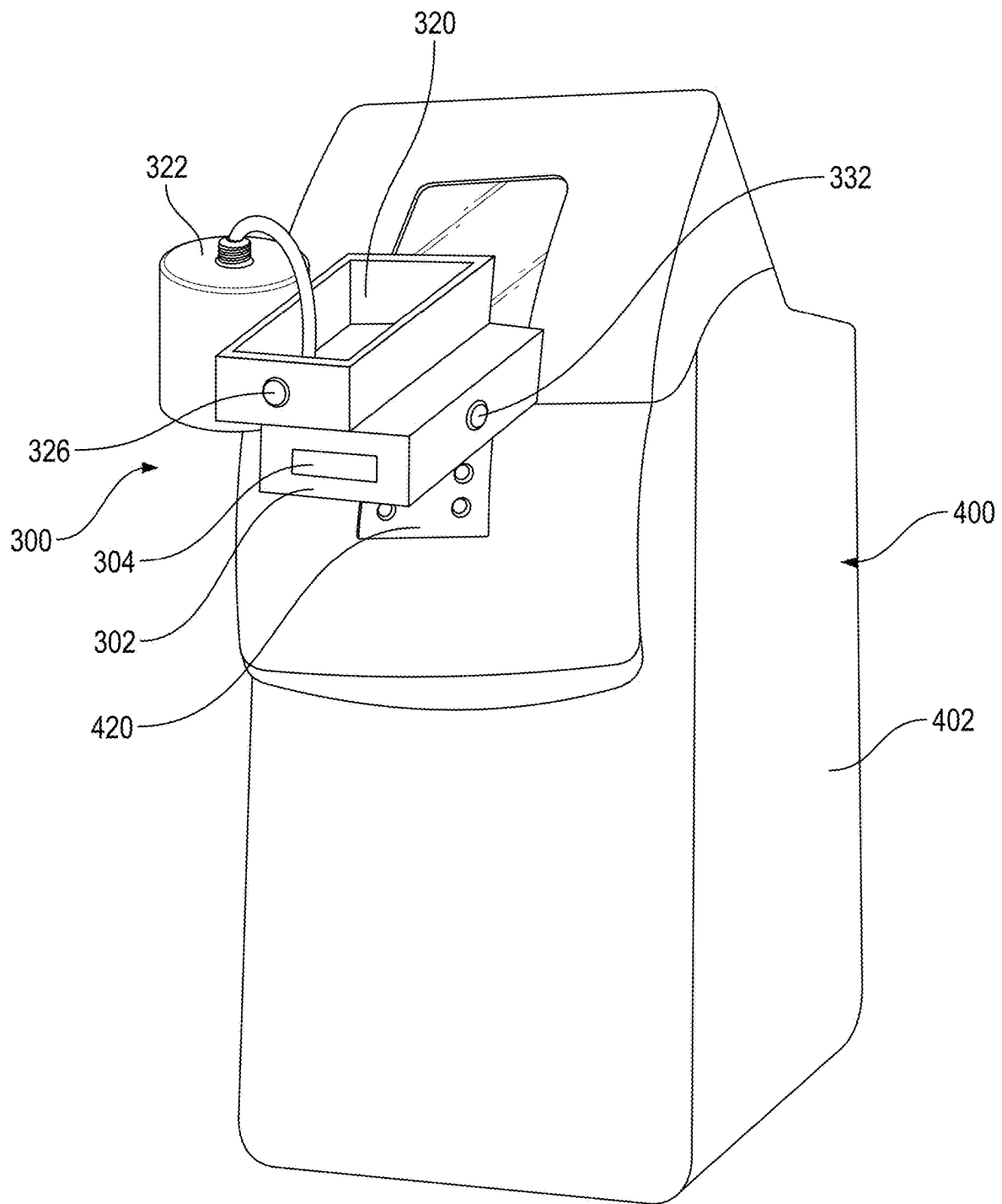


FIG. 21

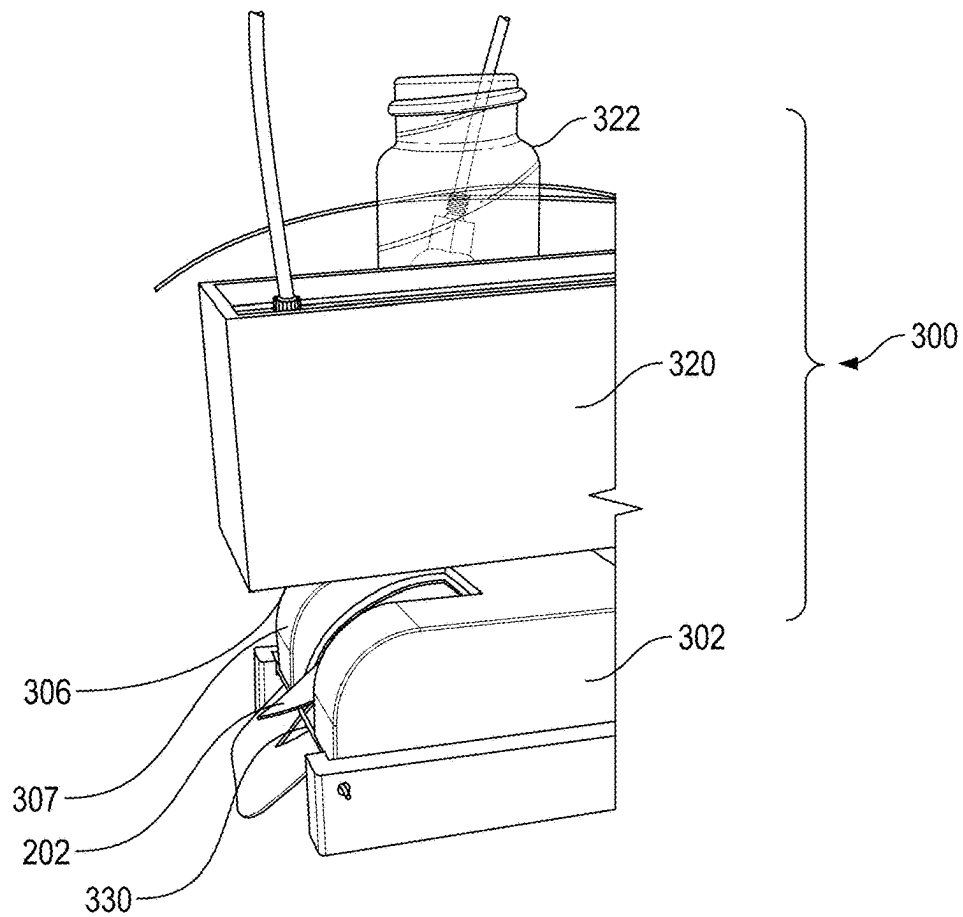


FIG. 22

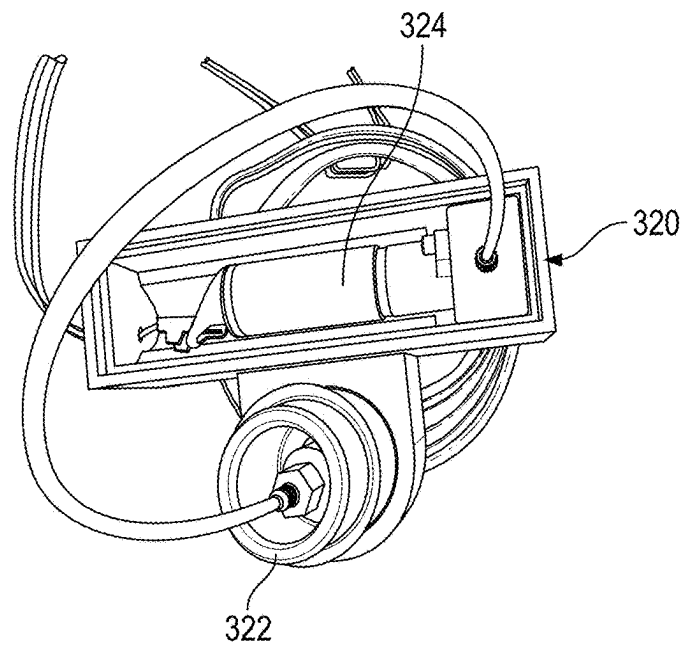


FIG. 23

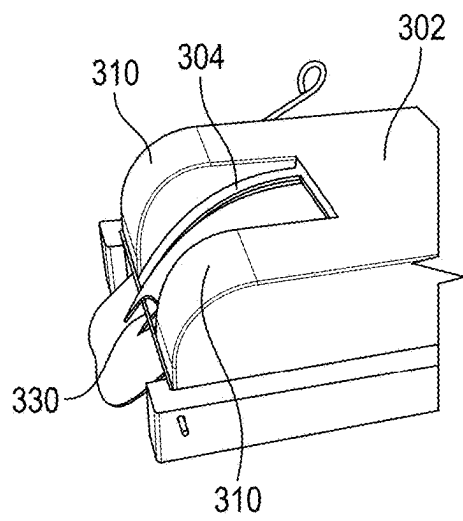


FIG. 24A

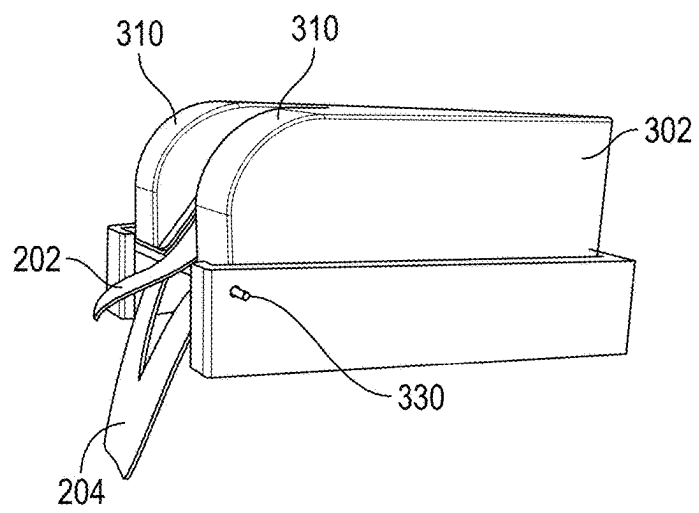


FIG. 24B

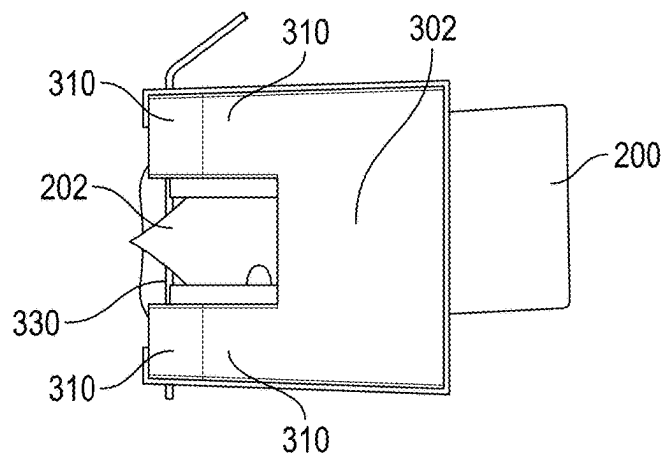


FIG. 24C

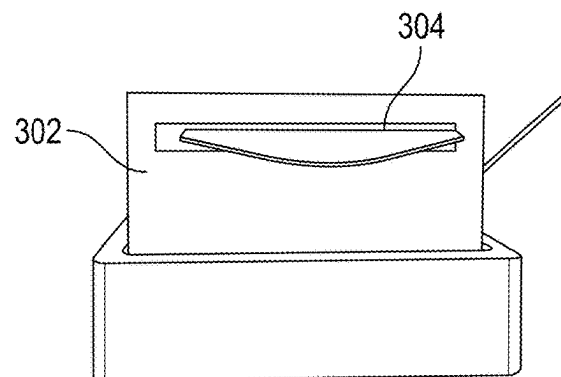


FIG. 24D

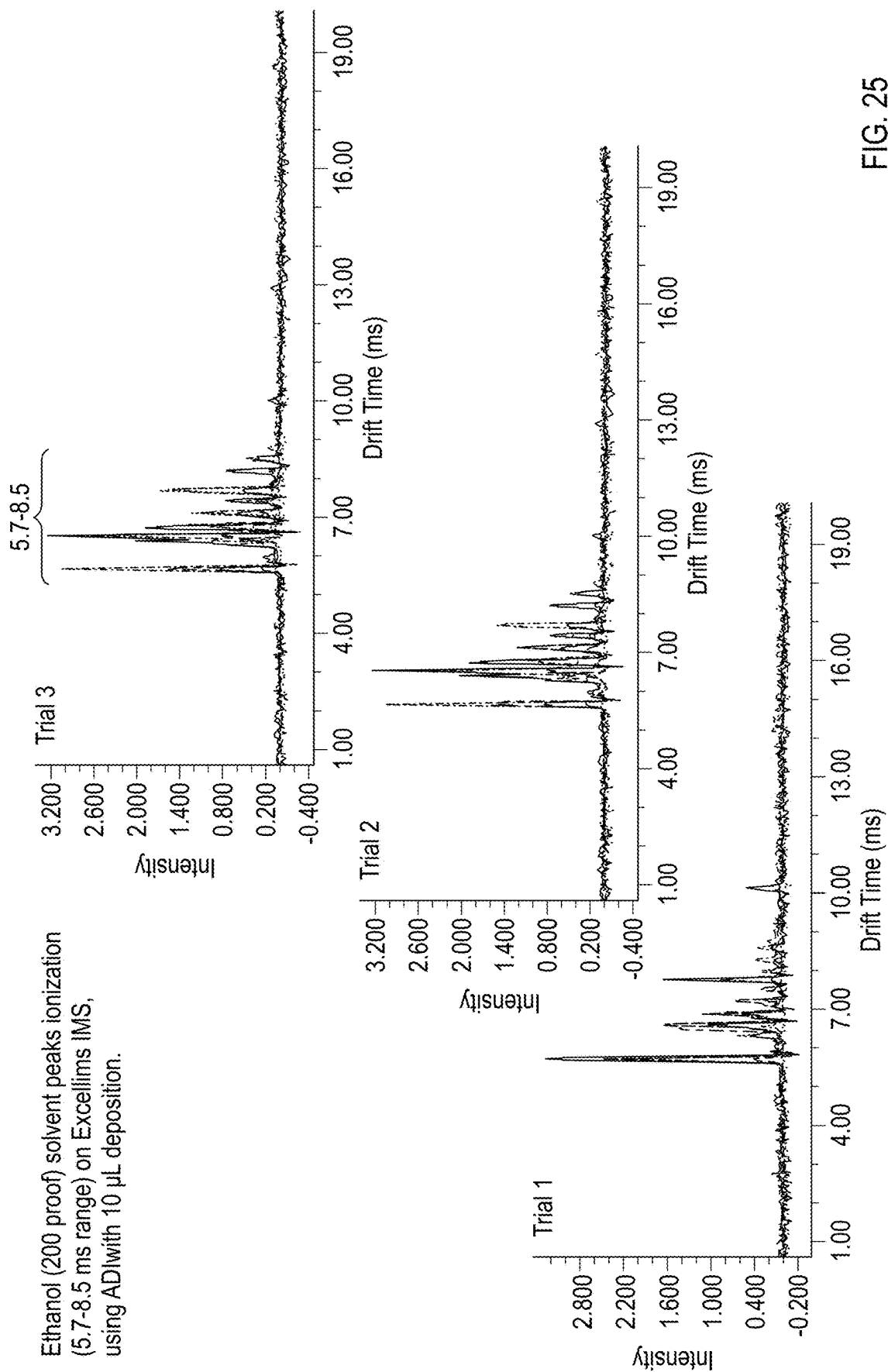


FIG. 25

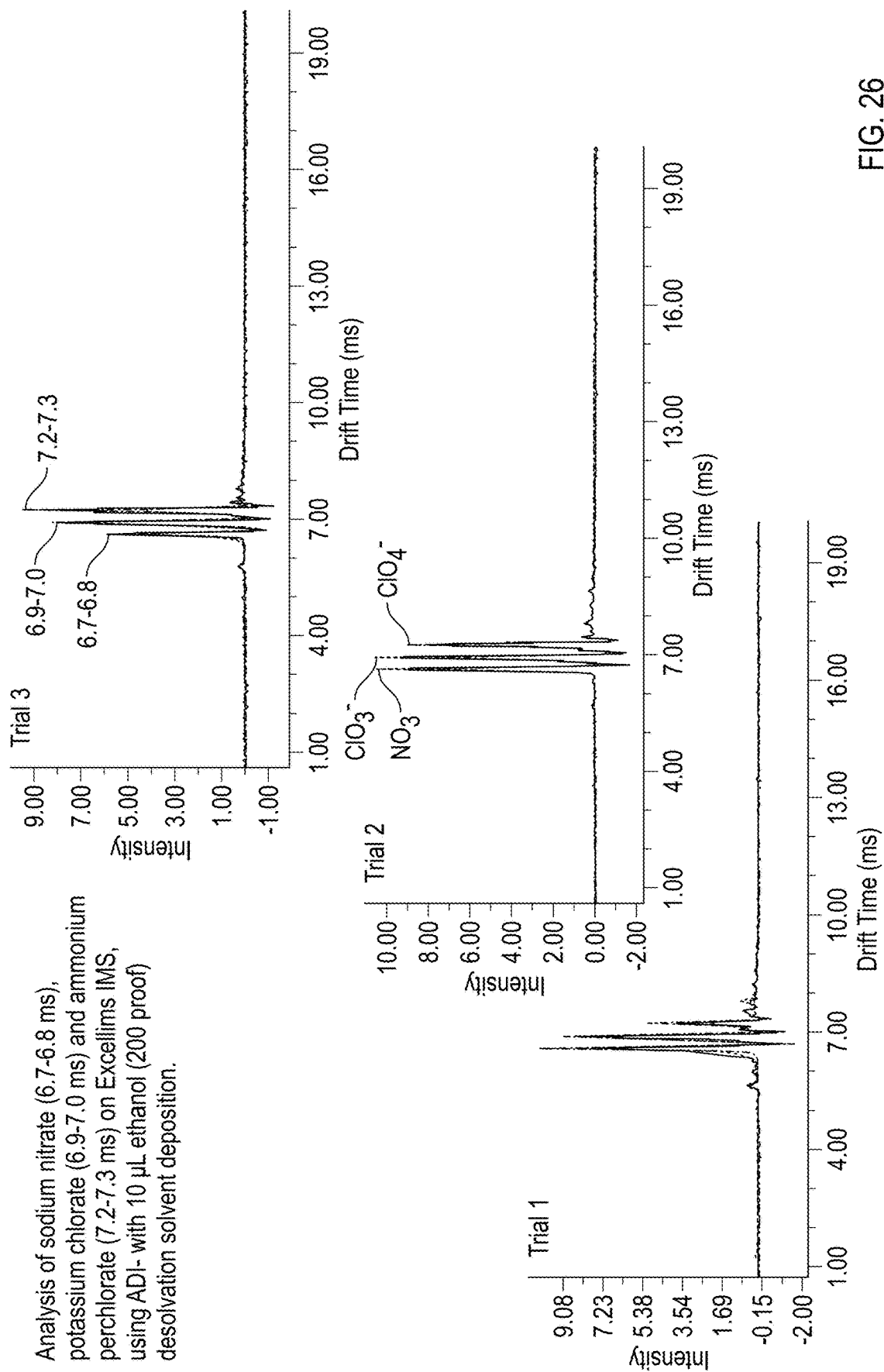
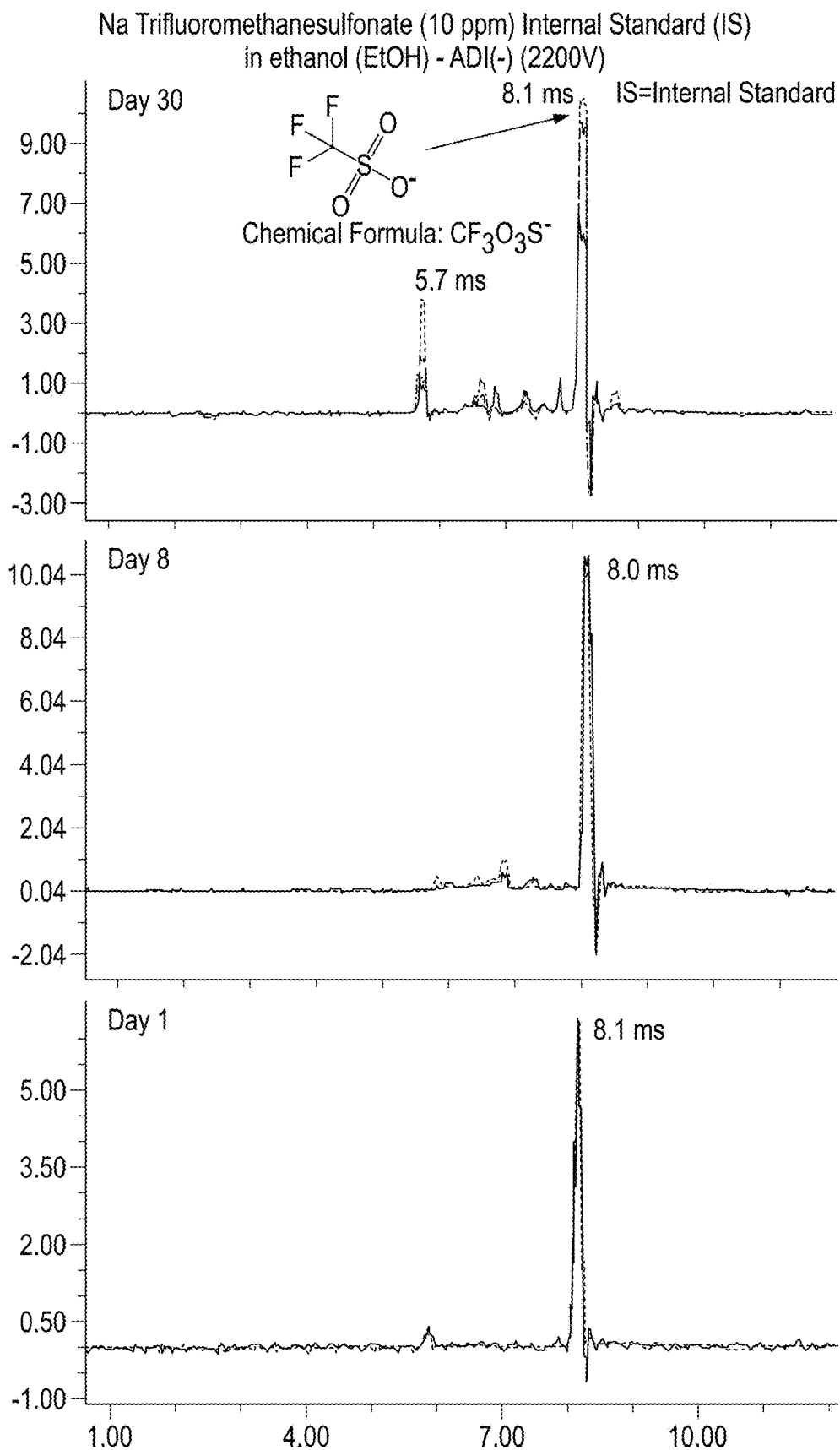


FIG. 26



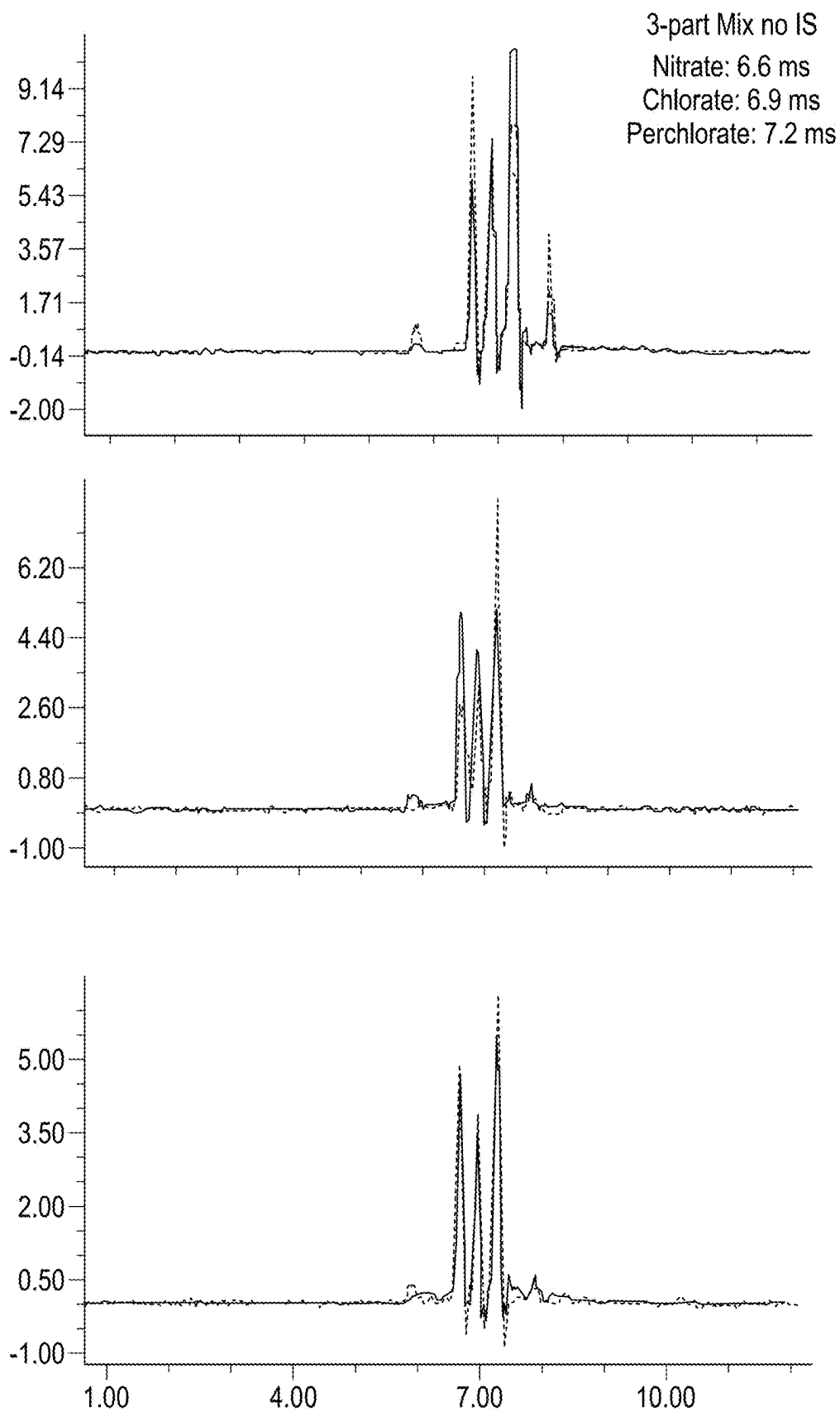
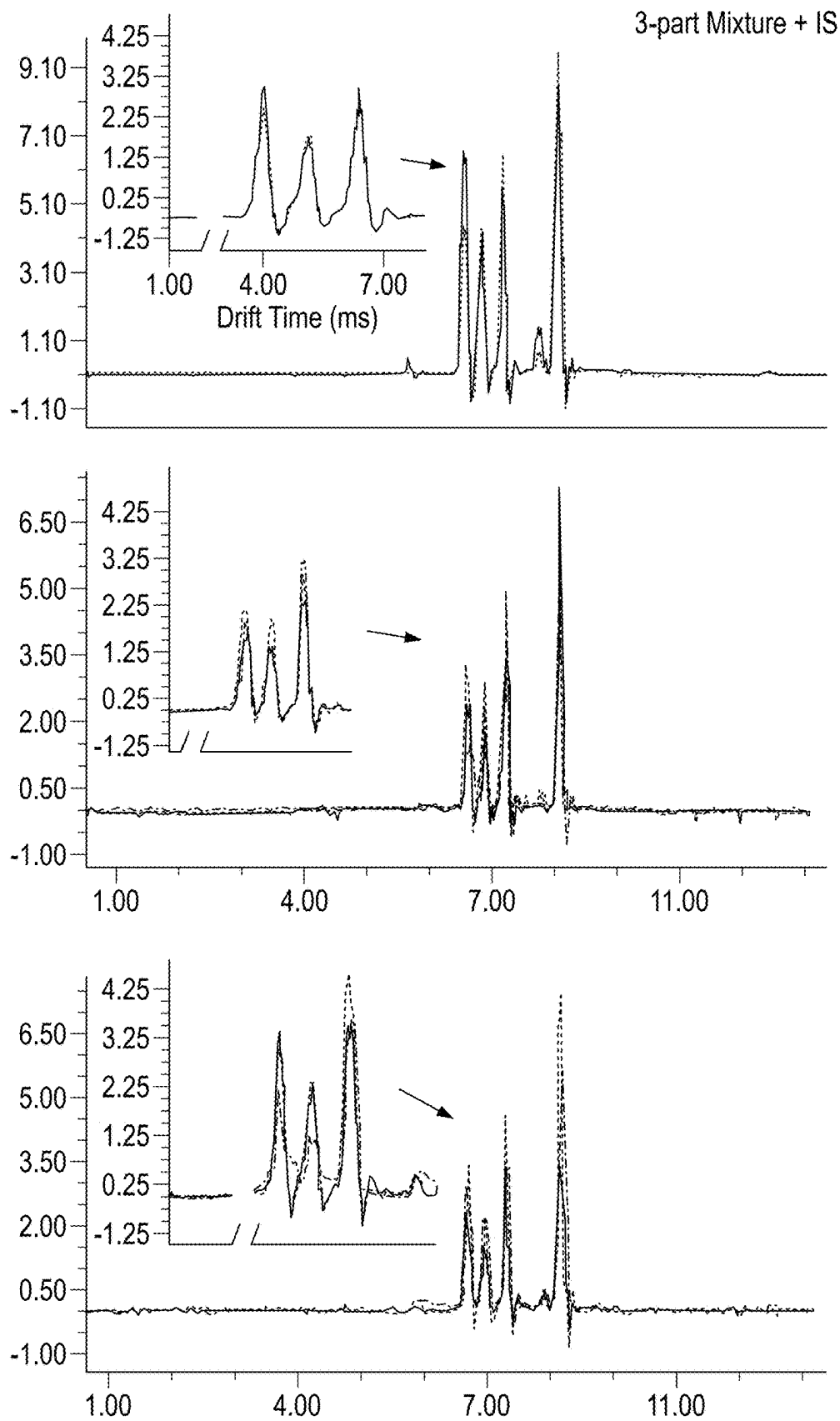
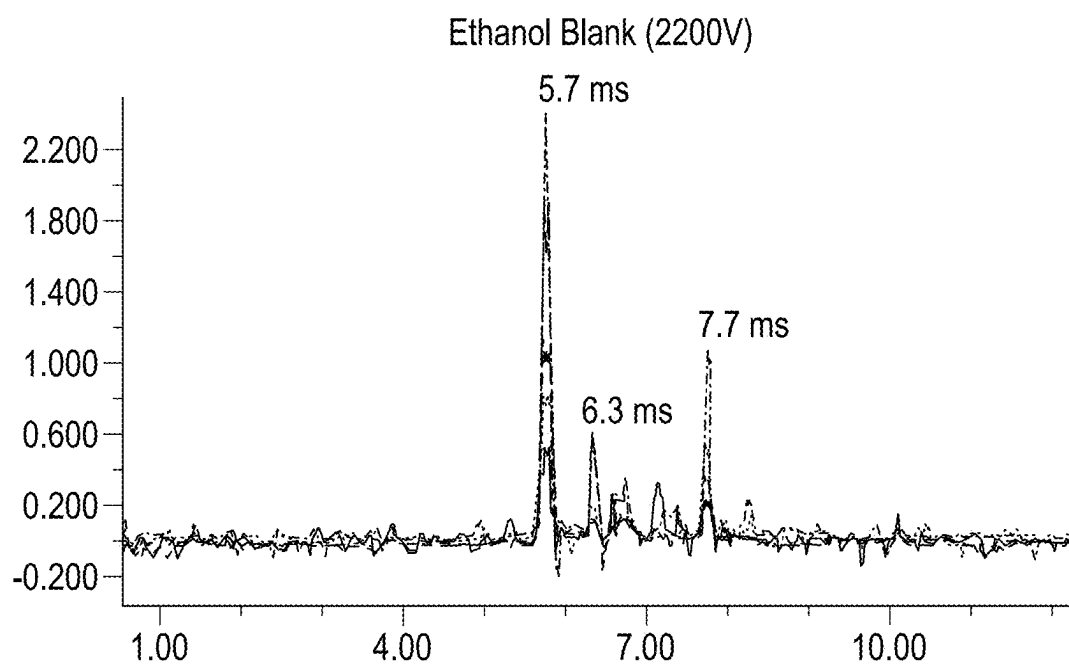


FIG. 27 (Cont.)





IS reduces solvent peaks & is in a region which won't interfere with
Organic Explosives: 9.5-13.8 ms Drugs: 11.6-12.1 ms

FIG. 27 (Cont.)

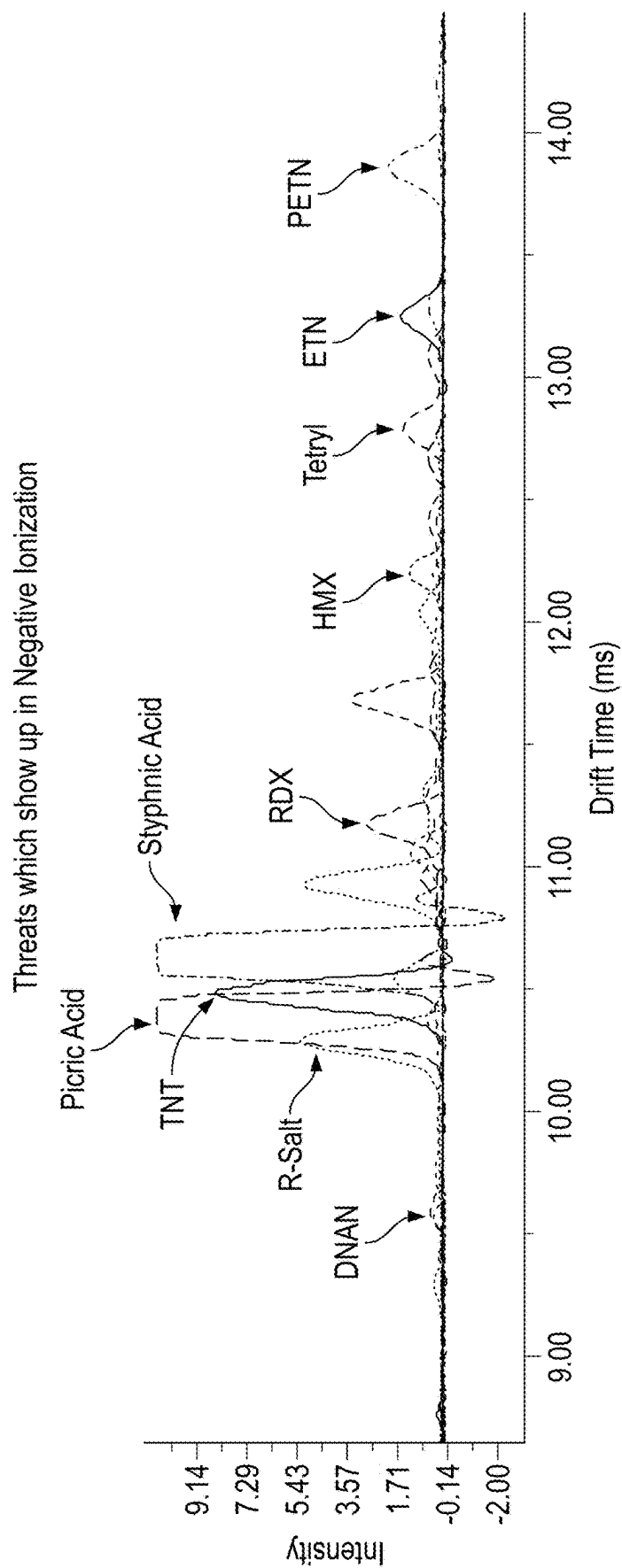


FIG. 28

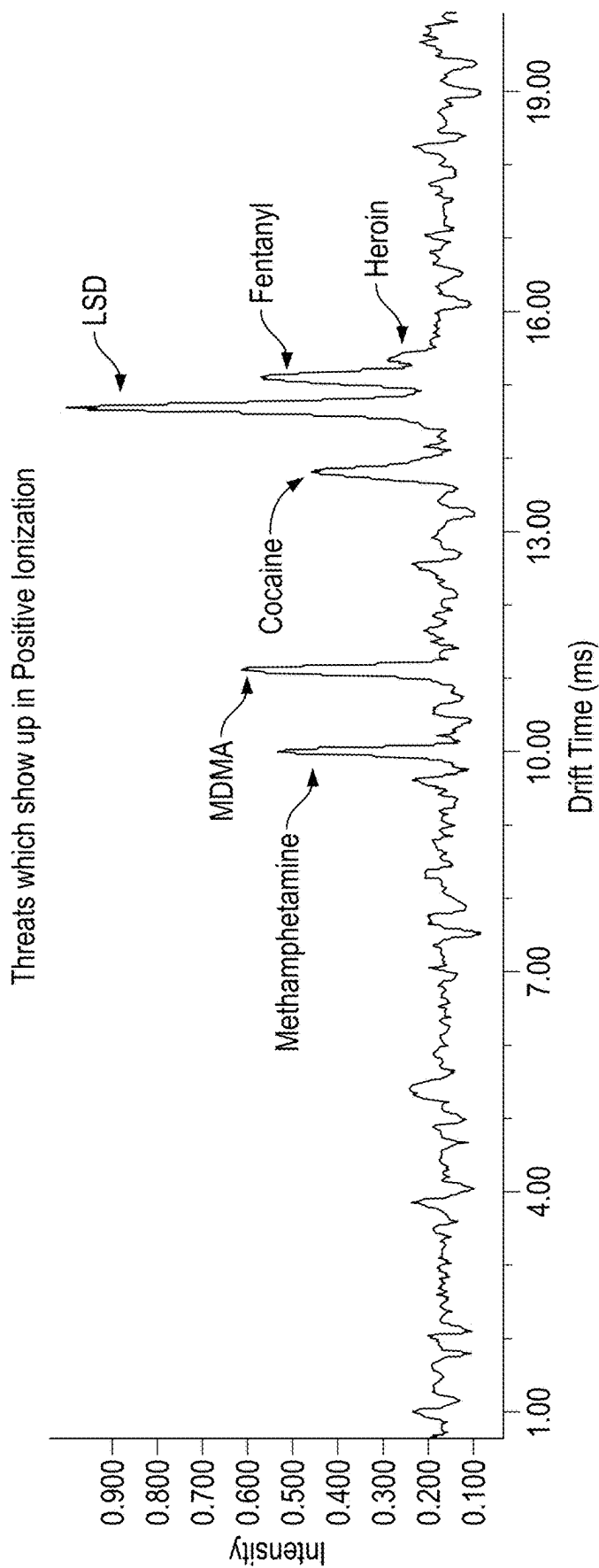


FIG. 29

APPARATUS AND METHODS FOR PREPARATION AND INTRODUCTION OF TRACE SAMPLES INTO AN IONIZING DETECTION SYSTEM

CROSS-REFERENCE TO RELATED APPLICATIONS

This application is a continuation-in-part of U.S. application Ser. No. 16/804,787 filed Feb. 28, 2020 which is related to and claims benefit of U.S. Provisional Application No. 62/816,253 filed Mar. 11, 2019, the entire contents of which are incorporated herein by reference.

STATEMENT OF GOVERNMENT INTEREST

This invention was made with government support under grant numbers HSHQDC-15-C-B0051 and HSHQDC-16-A-B0008 awarded by the DHS/ST. The government has certain rights in the invention.

BACKGROUND OF THE DISCLOSURE

The present invention generally relates to chemical detection systems and more specifically to illicit and dangerous substance trace detection systems, which are used in airports and other facilities for screening.

Ion Mobility Spectrometry (IMS) and Mass Spectrometry (MS) are known techniques for chemical detection.

Ion Mobility Spectrometry (IMS) is an atmospheric pressure chemical detection technology that is, in many ways, similar to Time-of-Flight Mass Spectrometry. Samples need to be in a gas phase for detection with an IMS system. Vaporized samples containing the analyte of interest are analyzed directly by drawing them into the IMS air flow system. Trace levels of explosive materials, drugs and chemical warfare agents can all be detected in this way. In existing IMS systems, samples are driven from a substrate or swab by applying heat from a desorber. Thus, non-volatile materials cannot be detected. Heat results in their decomposition rather than their desorption.

Sample material released into the IMS instrument by thermal desorption is then ionized to form reactant ion species from the gas employed in the system, normally air. Mixing these stable reactant ion clusters with vapor samples to be analyzed can result in ionization of the sampled materials, thus forming ion clusters characteristic of the sample material. A small cloud of formed ions is guided by an electromagnetic field through a drift tube towards a detector, where they eventually strike a collector electrode to produce a signal. Ions travel through the drift tube guided by the electromagnetic field and opposed by a gas counter-current. As a result, characteristic clusters of ions are formed and separated based on analytes' size, shape and cross-sectional area, which in turn produce specific drift speeds that can be further correlated to a calibration standard to create a final plasmogram result.

FIG. 1 illustrates, as prior art, an early portable IMS instrument generally indicated at 10. A swab 12, which has sampled an area of interest, is held in front of the inlet slot with a holder 14. A desorber heater heats the air around the swab or the swab itself to release volatile trace chemicals from the sample swab. Once inside the instrument 10, the gas analyte is ionized and passes through various magnetic grids to the collector/detector.

Mass Spectrometry (MS) is a similar analytical technique because it requires ionization of chemical species, but unlike

IMS it sorts the ions based on their mass-to-charge ratio. Another difference of MS is their requirement to operate under low vacuum conditions ($\sim 10^{-3}$ - 10^{-11} bar), unlike stand-alone IMS detection systems that work under atmospheric pressure conditions (i.e., ~ 1 bar). There are IMS devices that were integrated with MS instruments as hybrid or tandem designs and operate at lower pressures, but they lose their stand-alone functionality (i.e., being hand-held or easily portable, not requiring pumps and specialty gases, mobile, simple for user operation and accessible). These IMSs are no longer viewed as its own device, since it is an extension of MS and serves purely as a chromatographic (i.e., separation) addition prior to mass spectrometric (mass-to-charge) detection, with no ability of performing detection on its own. Mass spectrometry is used in many different fields and is applied to pure samples as well as complex mixtures.

FIG. 2 is a schematic of an exemplary magnetic sector mass spectrometer (MS) instrument generally indicated at 20. In a typical MS testing procedure, a sample, which may be solid, liquid, or gas is first converted to vapor (operation not shown on schematic). The vapor 22 is ionized in an ionization region 24, for example by a corona discharge needle. This may cause some of the molecules in the sample to break into charged fragments. These ions are then separated according to their mass-to-charge ratio, typically by driving them in an acceleration region 26 and then subjecting them to an electromagnetic field 28. Ions of the same mass-to-charge ratio will undergo the same amount of deflection by the electromagnetic field 28. The ions are counted by a detector 30, such as an electron multiplier. Results are displayed as spectra of the relative abundance of detected ions as a function of the mass-to-charge ratio. The original structure of the analyte can be elucidated by the observed mass-to-charge ratio and characteristic fragmentation pattern.

For a number of different purposes, including environmental testing, law enforcement, security and field detection, samples requiring MS or IMS analysis are collected on swabs.

FIGS. 3a-3c illustrate a prior art rectangular swab 50 and holder 52 and introduction of the swab 50 into an exemplary prior art IMS instrument 54 for testing. At the present time, swabs collected for travel security screening are tested directly without further sample preparation primarily because of time and technique constraints. This is a balance between adequate security and the need to move people through checkpoints at a reasonable pace.

FIG. 4 illustrates another prior art swab 60 and holder 62.

FIG. 5 still further illustrates additional swab shapes 70, 80, 90 currently being used in industry.

For obvious reasons, the reliability and speed of testing of the collected swabs is critical, for particularly for security screening at travel checkpoints, i.e. airports, cruise and border crossing. Accordingly, there is a perceived need for enhancement of the sample preparation to improve testing results without sacrificing testing speed.

SUMMARY OF THE INVENTION

The present disclosure teaches a novel sample preparation technique and provides a swab holder/positioner which aids in the delivery of the analyte into the instrument inlet and concurrently serves as an ionization source.

The technique includes adding a solvent to the swab to better dissolve and release any trace chemicals contained on the swab, charging of the solvated swab to create an ionized

analyte prior to entry into the detection instrument, and positioning of the charged swab so that a sharp corner, or a shaped angled tip of the swab faces the inlet into the detection instrument. The technique may be referred to as Ambient Desorption Ionization (ADD).

In some embodiments, buffers and an internal standard are added to the solvent to normalize detection peaks and calibrate the detection scheme in both MS and IMS.

In some embodiments where the swab includes pre-existing corners, i.e. 90 degree square corners, or otherwise, the swab can be used without modification (see swabs 50, 70, 80, 90 as illustrated in FIGS. 3a and 5).

As a first step in the technique, the swab is positioned so that an optimally angled corner or tip (analyte release area) faces the instrument inlet port to aid in creation of a directed ionized plume at the inlet of the detector. The swab is preferably held in an adapter/holder which is integrated into, or is configured for mounting onto, or positioning in front of, an existing MS/IMS detector device so that the corner or angled tip is within ± 5 cm of the entrance to detector. The adapter positions the swab and provides the high voltage connection. A solvent supply for the technique may also be supported by the adapter/holder.

The sample swab is energized by applying a high voltage to ionize the analyte and create ions which are directed toward the detector inlet. The detector inlet is grounded or has a differential potential in order to attract the ionized species into the detector inlet. The existing detector units thereafter function as intended for analysis.

In the alternative, if the swab lacks a pre-existing corner, such as the round swab 60 illustrated in FIG. 4, it can be quickly modified by cutting or otherwise shaping the swab. A shaping punch tool can be provided to quickly cut the swab 60 and create the desired angled tip or corner.

Additionally, some embodiments may utilize a swab with a pre-formed triangular analyte release area. The pre-formed area may be perforated or otherwise cut to provide the required triangular apex release shape.

The novel technique is thus intended to improve release of chemical residue from the sample swabs and improve detection. Sample introduction and ionization are combined in a single step process making this invention particularly useful for MS/IMS, since this new technique allows nonvolatile compounds to enter and flow through the instrument.

The invention advantageously allows the use of existing swabs directly as the ionization source, given the availability of a high voltage supply internal or external, and eliminates the need for other types of ionization sources, for example radioactive material or corona discharge.

The ability to use present swab materials and holders is critically important in large scale deployment of the technique, and depending on the IMS or MS device, the present technique and adapter may require no change in swab materials; and if the instrument is presently portable, it remains so.

BRIEF DESCRIPTION OF THE DRAWINGS

While the specification concludes with claims particularly pointing out and distinctly claiming particular embodiments of the instant invention, various embodiments of the invention can be more readily understood and appreciated from the following descriptions of various embodiments of the invention when read in conjunction with the accompanying drawings in which:

FIG. 1 illustrates an exemplary prior art Ion Mobility Spectrometer (IMS);

FIG. 2 illustrates an exemplary prior art magnetic sector Mass Spectrometer (MS);

FIGS. 3a-3c illustrate a prior art sample swab (rectangular), wand and detector instrument;

FIG. 4 illustrate a round or circular prior art swab and wand;

FIGS. 5a-5c illustrate additional prior art swabs which have angled corners 90-120 degrees;

FIG. 6 schematically illustrates the preparation technique and release of the analyte from the sample in accordance with the present disclosure using an unmodified swab;

FIG. 7 illustrates an exemplary testing instrument including the present sample preparation apparatus;

FIGS. 8a-8c schematically illustrate exemplary positioning angles for holding the sample swab in front of the inlet port of the testing instrument;

FIG. 9 illustrates a circular sample swabs and an exemplary stamp or punch which is configured to cut/shape the circular swab to provide it with an analyte release tip having preferred release angle;

FIG. 10 is another schematic illustration of the exemplary testing instrument with the shaped swab as illustrated in FIG. 9;

FIG. 11 illustrates an exemplary holder or adapter for positioning of the swab or shaped swab and for application of the solvent to the swab;

FIGS. 12a-12d illustrate test data showing the same or similar performance between the present ADI technique and an ElectroSpray Ionization (ESI) technique on an LTQ-Orbitrap instrument;

FIGS. 13a-13d illustrate additional test data showing the same or similar performance between the present ADI technique and ESI on an LTQ-Orbitrap instrument;

FIG. 14 illustrates examples test results of explosive detection using ADI on an Explosive Trace Detector (ETD) Device;

FIG. 15 illustrates examples test results of explosive detection using ADI versus desorption on a standard ETD Device;

FIGS. 16a-16d illustrate examples of detection of Nitrate-, Perchlorate-, Chlorate-Salts and TNT on an Excellims IMS Device with MS Confirmation;

FIG. 17 is a plan view of another exemplary pre-formed sampling swab;

FIG. 18 is a perspective view thereof;

FIG. 19 is a schematic illustration of the exemplary swab of FIGS. 17 and 18 deployed with a novel swab holder, positioner and voltage supply arrangement;

FIGS. 20A-D are various view of the swab holder and positioner with the integrated voltage delivery contact positioned below the swab;

FIG. 21 illustrates an exemplary compact Mass Spectrometer (MS) device with the swab holder/positioner and integrated solvent dispenser mounted to an inlet port of the MS device;

FIG. 22 is a side view of the swab holder and integrated solvent dispenser;

FIG. 23 is a top view thereof;

FIGS. 24A-D are various view of the swab holder/positioner showing deflection of the flanking side portions of the preformed swab and presentation of the apex of the analyte sensing area;

FIG. 25 illustrates ethanol (200 proof) solvent peaks (among many other solvents) ionization (5.7-8.5 ms range) on Excellims IMS, using ADI—with 10 μ l deposition;

FIG. 26 illustrates and analysis of sodium nitrate (6.7-6.8 ms), potassium chlorate (6.9-7.0 ms) and ammonium per-

5

chlorate (7.2-7.3 ms) on Excellims IMS, using ADI—with 10 μ L ethanol (200 proof) desolvation solvent deposition;

FIG. 27 illustrates a side-by-side analysis of an ethanol buffered solvent with and without an internal standard;

FIG. 28 illustrates peak analysis of potential threats which show up in negative ionization; and

FIG. 29 illustrates peak analysis of potential threats which show up in positive ionization.

DETAILED DESCRIPTION OF THE EXEMPLARY EMBODIMENTS

Certain exemplary embodiments will now be described to provide an overall understanding of the principles of the structure, function, manufacture, and use of the device and methods disclosed herein. One or more examples of these embodiments are illustrated in the accompanying drawings. Those skilled in the art will understand that the devices and methods specifically described herein and illustrated in the accompanying drawings are non-limiting exemplary embodiments and that the scope of the present invention is defined solely by the claims. The features illustrated or described in connection with one exemplary embodiment may be combined with the features of other embodiments. Such modifications and variations are intended to be included within the scope of the present disclosure. Further, in the present disclosure, like-numbered components of the embodiments may have similar features, and thus within a particular embodiment each feature of each like-numbered component is not necessarily fully elaborated upon. Additionally, to the extent that linear or circular dimensions are used in the description of the disclosed systems, devices, and methods, such dimensions are not intended to limit the types of shapes that can be used in conjunction with such systems, devices, and methods. A person skilled in the art will recognize that an equivalent to such linear and circular dimensions can easily be determined for any geometric shape. Further, to the extent that directional terms like proximal, distal, top, bottom, up, or down are used, they are not intended to limit the systems, devices, and methods disclosed herein. A person skilled in the art will recognize that these terms are merely relative to the system and device being discussed and are not universal.

Referring now to the drawings, exemplary embodiments of the invention are generally described and illustrated in the attached figures. The present disclosure teaches a novel sample introduction technique and provides an adapter/holder which aids the delivery of the analyte into the testing instrument inlet and concurrently can serve as an ionization source. The technique includes adding a solvent, charging the swab to create an ionized analyte plume, which, due to positioning of the swab corner or shaping of the swab, more efficiently directs the ion plume into the entry of the detection device.

Referring now to FIGS. 6-11, the technique and adapter/holder apparatus are illustrated in accordance with the invention. A swab is generally indicated at 100 and includes an optimally angled analyte release area 102 having an apex angle measuring from about 30 degrees to about 120 degrees. The illustrated example is a conventional rectangular swab having a distal end with a pre-existing corner angle of about 90 degrees. Any existing swab material may be utilized, although materials with a rougher surfaces, grooves or channels may produce improved results. A particularly suitable sampling swab material is described in U.S. Pat. No. 9,200,992, the entire contents of which are incorporated herein by reference.

6

In use, the swab 100 will be contacted with an object to be tested and will thereafter carry a trace chemical residue 104.

An MS or IMS instrument is provided and generally indicated at 106. The MS/IMS instrument 106 is conventional in the art and generally operational as described hereinabove. The instrument 106 has a housing 108 and an inlet port 110. Generally described, to create an ionized plume of analyte directed into the inlet 110 of the instrument 106, an optimally angled corner or tip (analyte release area) 102 of the sample swab 100 is pointed at and preferably positioned within ± 5 cm of the inlet port 110 of the instrument 106.

A solvent supply is provided and generally indicated at 112. The solvent supply 112 includes a container 114 for holding the solvent 116 and a dispenser 118 which may be advantageously positioned to selectively apply solvent 116 to the swab 100.

Solvents 116 may include water, methanol, ethanol, isopropanol or other alcohols, acetonitrile, acetone, benzene, hexane, ethyl acetate, DMF, THF, a combination of solvents, or aqueous buffers. Ethanol may be considered a preferred solvent.

Use of an aqueous buffer allows introduction of potential adduct forming species or dopants. Examples of such species include chloride, formate, acetate, ammonium, sodium, potassium, nitrate.

A high voltage source 120 is provided for charging of the swab 100 and generally includes a ground 122. Where necessary for safety the ground can be provided by connection to the instrument inlet port 110. A power contact (+/-) 124 which is connected to the swab 100 is essential to the technique. In this regard, in order to grab or hold the swab 100 and provide the needed charge adjacent to the analyte release area 102, the apparatus may include a contact clip 126 which may have spring fingers (as illustrated) which frictionally grasp the swab 100. The clip 126 may also comprise a spring-loaded clamp or clip device. The contact clip 126 is preferably a disposable component which can be discarded after each use to prevent cross-contamination from one sampling swab to another. In this regard, the body of the clip is releasably connected with the power contact 124, in a sliding socket 128 (as illustrated), or otherwise as known in the connector art.

Turning now to FIG. 7, the swab 100 is mounted in/positioned on an adapter/holder 130 which is configured for mounting onto the exterior housing 108 of an existing detector-device 106. In the illustrated example, the holder 130 is received on the housing 108 of the MS/IMS device 106 and provides a platform 132 on which the sampling swab 100 can be positioned adjacent to the inlet port 110. The holder 130 has an adapter 134 (spring clip) which may frictionally hold the swab 100 in a desired position. Alternately, as seen in FIG. 11, the adapter 134 may be provided in the form of a plastic molded holder into which the swab 100 is positioned and held. The adapter 134 may then be snap received or otherwise removably positioned or mounted onto the platform 132. The adapter 134 may also provide a solvent guide port 136 which may be connected with the solvent supply dispenser 118.

Referring briefly to FIG. 8, the inlet port 110 has a defined entry plane (FIG. 8a) A and as earlier noted, the ± 5 cm positioning is defined as the tip area 102 being positioned somewhere between 5 cm inside the inlet port 110 or within 5 cm in front of the inlet port 110. FIGS. 8b and 8c also illustrate that the platform 132 of the holder 130 can be

angled or tilted 0-30 degrees upward or downward or to the left or right to provide improved gravitational disbursement of the solvent **116**.

The swab holder **130** allows support of the swab **100** and direct application of an adjustable voltage (0.5V to 50 kV but less than 10 kV of floating voltage is preferred) directly behind or on or slightly above the solvated sweet spot to ionize the analyte and create an ion plume **140** directed toward the detector inlet (See FIG. **6**). The voltage supply **120** may be provided with a switch **142**, which may be on the holder **130**, or elsewhere, to selectively apply the voltage as desired. As noted above, the inlet port **110** is grounded or held at a differential potential. The differential potential created between the swab **100** and the entrance **110** to the IMS or MS **106** acts as the equivalent of electrospray ionization and drives the analyte toward the detector inlet **110**.

Alternatively, the holder **130** may be configured for placement in front of the instrument **106** so that currently deployed detector units can still be utilized. Still further, the apparatus as described herein may be fully integrated with the MS/IMS instrument for a complete integrated instrument.

Turning to FIG. **9**, if the swab has no corner, such as the circular swab illustrated in FIGS. **4** and **9**, then the swab **100A** can be cut or otherwise shaped to form an optimally angled tip area **102A**. An exemplary angle is approximately 45 degrees but may be less or greater ranging from 30 degrees (See FIG. **11**) up to and including about 120° as desired. The release area **102** may also be slightly truncated or flattened. Shaping may be accomplished by any number of means, such as manual scissors, or an external punch device **144**, folding, or a pre-perforated swab (not shown) may be provided which is separated along the perforations, and/or a premade channel to direct charge and solvent.

Exemplary test data for the present ADI sampling and preparation method are illustrated in FIGS. **12-16**. The proposed ADI technique was evaluated on commercially available instruments—Thermo LTQ Orbitrap XL mass spectrometer and Excellims attached to a Thermo Exactive mass spectrometer. The samples were prepared from neat materials by diluting them in appropriate solvent (e.g. acetonitrile, methanol, ethanol, isopropanol, water, but any other solvent can be used). For sample introduction, a direct infusion into the MS was employed. Resulting spectra was collected and averaged over at least 25 scans (acceptable number of scans for statistical verification of sample average response). The precursor ion was isolated and resonance energy was increased (CID in eV) until characteristic products were produced from corresponding ion. ADI produced virtually identical spectra to that of established ESI technique. Additional parameters were tested to verify the versatility of the proposed innovation. Many compounds do not ionize on their own, but can accept nearby charged species as carriers, thus forming what is known as adducts. ADI was tested on whether most common adducts can be produced (e.g. chloride, acetate, formate, ammonium, sodium, etc.) and produced spectra was compared to prior art ESI technique. The adduct was introduced directly by dissolving ammonium chloride (200 µM) in the solvent system designed for delivery. (see FIGS. **12** and **13**).

The proposed technique was further tested on commercially available IMS systems to verify its applicability. The ADI source was positioned directly at the inlet of the Ion Track IMS. The power source was attached directly to the swab, which was manipulated (positioned or shaped) so that a point was directed toward the inlet of the instrument, and

the voltage was controlled through the charger interface (3-20 kV). Once the sample was ionized and grounded to the entrance of the IMS, the scan initiation was manually triggered to acquire the chromatogram (FIGS. **14** and **15**). To ensure there was no carry over or false positive results, the system was first cleared out, and then a blank sample was run that contained only solvent system but no analyte immediately followed by ADI introduction technique with analyte present.

ADI was further tested by coupling it to IMS (Excellims)/MS (Thermo Exactive high accuracy mass spectrometer) system. This setup gives the ultimate confirmation of 1) ionization efficiency, 2) analytes' true identity via accurate mass confirmation, and 3) robustness of the ADI via semi-quantification mechanisms. The setup provided unequivocal confirmation of ADI source suitability for coupling with IMS/MS systems for which it was originally intended for. The presence of non-volatile compounds corresponding to m/z of nitrates, perchlorates and chlorates in the spectra of MS (FIG. **16**) upon selection of the peaks in IMS confirmed that ADI can move non-volatile compounds into a gas phase without their decomposition. Alternatively, organic compounds (e.g. TNT) can also be detected without visible degradation allowing for wider application of ADI source in comparison to thermal desorber.

Turning to FIGS. **17** and **18**, there is illustrated and disclosed a novel pre-configured shaped swab **200** which is generally rectangular in shape and may be provided with a triangular analyte release area **202** having a pointed tip with apex angle of between 30 and 120 degrees. The triangular analyte release area **202** may be pre-perforated in the body of the swab or pre-cut. The central analyte release area **202** is centered between flanking side portions **204** which serve to retain the general rectangular shape of the swab for better retention in a conventional swabbing wand. The flanking side portions **204** may be connected forward of the tip or may be separated. An exemplary apex angle is approximately 45 degrees but may be less or greater within the noted range.

As described above, the swab **200** may be rubbed across a surface to collect potential analytes onto the analyte release area **202**, or it could remain stationary having the potential analyte placed onto the release area **202**. In either case, the swab **200** containing the potential analyte must be presented to the detection instrument, and for this a holder is required.

Referring to FIGS. **19** and **20A-D**, the apex tip shape of the swab **200** is important for creation of a "Taylor" cone of ions directed into the inlet **410** of the detection device **400** (See FIG. **19**). The above-described swab **200** fits into a holder **300** which requires no support of the swab **200** by the user during use. FIGS. **19** and **20A-D** illustrate an exemplary configuration of a swab holder **300** comprising a body or housing **302** with a swab positioning slot **304** for receiving a swab **200**, a dispensing nozzle **306** for dispensing a solvent **307**, and a voltage delivery contact **308**. The housing **302** is configured with deflection arms **310** to receive the pre-configured swab **200** and deflect the flanking side portions **204** downwardly out of the way such that the pointed release area **202** can be positioned immediately adjacent to and directed towards the grounded detector inlet **410** or entrance. The voltage delivery contact **308** is configured and mounted within the housing **302** between the deflector arms **310** such that the analyte release area **202** is engaged with the contact **308** when fully inserted into the slot **302** and properly positioned and spaced at the detector entrance **410**.

Referring to FIGS. 21 through 24D there is illustrated an MS or IMS instrument generally indicated at 400 which incorporates the novel swab holder positioner apparatus 300 as described above. In some embodiments, the swab holder/positioner apparatus 300 is configured to be secured to the device housing with brackets 420 or other fasteners which properly align the device 300 for presentation of the analyte release area with the inlet 410.

The MS/IMS instrument 400 may be conventional in the art and generally operational as described hereinabove. While the above embodiments were primarily described in connection with an exemplary IMS device, the present configuration is an exemplary MS testing device, such as an Advion™ ExpressionCMS™ (Comact Mass Spectrometer). Other MS type devices are also contemplated within the scope of the present disclosure.

The instrument 400 has a housing 402 and an inlet port 410. Generally, as described, to create an ionized Taylor Cone plume of analyte directed into the inlet 410, the tip (analyte release area 202) of the swab 200 is pointed at and may be positioned within 5 cm of the inlet port 410. In some embodiments, the distance may be varies and the tip 202 may be positioned slightly above the inlet axis.

The swab holder/positioner apparatus 300 generally comprises a housing 302 with a swab positioning slot 304 extending through the housing for receiving swab 200, a solvent dispenser 320 including a solvent supply container 322 containing a solvent 307, a solvent dispenser nozzle 306 positioned adjacent to the analyte release area 302 and a pump 324 in fluid communication with the solvent supply container 322 and the solvent dispenser nozzle 306. The holder/positioner 300 may further include an integrated a high voltage supply 305 including delivery contact 308 and a ground communicating with the device inlet 410.

The housing 302 is configured with deflection arms 310 to receive the pre-configured swab 200 and deflect the flanking side portions 204 downwardly out of the way such that the pointed release area 202 can be positioned immediately adjacent to and directed towards the grounded detector inlet or entrance 410. The housing 302 may further be provided with a cross-bar or other deflection structure 330 extending across the slot 304 for deflecting the analyte release area upwardly for better presentation without user handling.

In another embodiment of the housing where the swab has been preshaped angles are already within the described angles 30-120 degree, in particular 90 degrees, the housing angles the swab so that the desired apex may be positioned within 5 cm of the inlet port 410.

The voltage delivery contact 308 is configured and mounted within the housing 302 between the deflector arms 310 such that the analyte release area 202 is engaged with the contact 308 when fully inserted into the slot 304 and properly positioned and spaced at the detector entrance 410. The apparatus allows support of the swab 200 and direct application of an adjustable voltage (0.5V to 50 kV but less than 10 kV of floating voltage is preferred) directly to the solvated analyte release area 202 to ionize the analyte and create an ion plume (Taylor Cone) directed toward the detector inlet 410. The voltage supply 305 may be provided with a switch 332 to selectively apply the voltage as desired. As noted above, the inlet port 410 is grounded or held at a differential potential. The differential potential created between the swab 200 and the entrance 410 to the MS acts as the equivalent of electrospray ionization and drives the analyte toward the detector inlet.

The voltage may be provided from above, below or on the side of the swab; however, it is generally, advantageous that

the voltage be applied on the opposite side of the swab as the application of solvent, but this is a space consideration, not a requirement of the structure or method.

The solvent supply pump 324 may be manually actuable or electrically powered and may be selectively actuable to dispense a predetermined volume of solvent 307 from the solvent supply container 322 through said dispensing nozzle 306 and onto the analyte release area 202. A small amount of solvent 307 aids the movement of the analyte across and off the matrix (swab) toward the inlet of the detection device. It is not desired that solvent enter the detection device; therefore, the solvent amount is kept small, e.g. 10 microliters, but may range from 1 microliter to 100 microliters. The solvent is expected to evaporate (vaporize) leaving charged analyte ions to enter the detection device. It may be necessary to keep the inlet area slightly warm and/or provide a gentle breeze to aid solvent evaporation.

The exemplary embodiment includes a commercial pump 324 which dispenses the desired amount of solvent, e.g. 10 microliters. The pump 324 may be positioned over the swab holder housing 302 as illustrated or beneath the housing 302 and pumped upwardly with a dispensing nozzle over the analyte release area. Positioning of the swab 202 within the holder slot 304 could trigger the release of the solvent or the solvent could be released by a separate switch/button 326. Applying the solvent via a wicking action either from the top or bottom of the swab is also feasible. Solvent should be applied either just behind or directly on the area where the sample is positioned on the swab.

Solvent Standard and Ionization Suppression

The ionization of analytes is a driving force behind the detection of the analyte in ion mobility spectrometry (IMS) and in mass spectrometry (MS). If ionization is somehow suppressed, then the analyte of interest produces little signal in the detector. Ionization suppression can happen because other species in sample (the matrix with the analyte) "steal" the charge. Ionization suppression often happens during analysis of complex matrices (e.g. blood, plasma, tissue, soil, run-off water, waste water, or in general any organic, inorganic, polymeric, natural and synthetic materials). Ionization enhancement can also happen, and is the opposite of ion suppression, i.e. observable signal intensity is increased. In either case, the minimization of these effects is preferable, especially for quantification and detection applications.

In ADI application, swabs (or any other sample introduction or separation platforms, e.g. TLC plates, 2D printed microchannel plates, etc.) are used as a platform for sample collection and/or introduction into the IMS/MS device. A solvent or a mixture of solvents is then used to solvate analytes off the surface of the collection platform. High voltage is supplied to analyte release area on the swab/platform where the solvent is then ionized and drives the analytes into the IMS or MS devices, while transferring its charge to the analyte.

The use of solvent in the ADI method introduces the possibility to predominantly ionize solvent and not analyte of interest. This can result in an ion suppression of the analyte. In IMS that uses drift time as the main metric for analytes discrimination, the solvent molecules can obscure detection of analytes if they have similar drift times.

Referring to FIG. 25 there is shown an ion mobility spectrometry trace of 200 proof ethanol. The response observed in the 5.7 to 8.5 ms range is from the solvent alone. All solvents potentially used in ADI produced some sort of response: methanol; acetonitrile, isopropyl alcohol, acetone; water; DMSO; methylene chloride; 50/50 ethanol/water. Such response from these various solvents could preclude

11

detection of analytes whose drift times overlap with solvent peaks. Their responses can be obscured or produce false positive results. In either case this is unwanted outcome for testing and detection purposes. However, it has been found that adding certain analyte species or the analyte itself to the solvent suppressed the ionization of the solvent, allowing the analytes (example nitrates, chlorates and perchlorates) as illustrated in FIG. 26, to be observed in a clean IMS trace.

Additionally, it has been discovered that addition of certain compounds can suppress ionization of solvent peaks, but not affect ionization of analytes. These “suppressant” standards can be directly added to the solvent or a mixture of solvents at parts-per-million levels or embedded into the sample collection/ionization platform. “Suppressants” can be tuned to cover specific drift time region and/or ionization potential (i.e. positive or negative). The same “suppressant” can be used for positive and negative ionization mode (assuming it is capable of both positive and negative ionizations in the first place).

To be introduced into the solvent system, the suppressant standards require multi-day stability in the solvent and must not, themselves, have drift times that overlap with the analyte(s) of interest.

Examples of such species include but is not limited to sulfates, sulfonates, phosphates, phosphides, chlorides, chlorates, perchlorates, nitrates, nitrites, and ammonium and phosphonium salts. Specific examples include but not limited to sodium dodecylbenzenesulfonate, 3-(1-pyridinio)-1-propanesulfonate, sodium diphenylamine-4-sulfonate; sodium trifluoromethane sulfonate, potassium phthalate monobasic, tetrabutylammonium phosphate; tetraphenylphosphonium chloride; benzylpyridinium salts.

These compounds serve two purposes, namely that of suppressing solvent signal (as discussed above), and that of providing a benchmark for the observed drift times so that daily instrument and ambient fluctuations can be accounted for.

FIG. 27 illustrates a side-by-side analysis of an ethanol buffered solvent with and without an internal standard and a comparison of multi-day stability within the solvent.

FIG. 28 illustrates peak analysis of potential threats which show up in negative ionization and FIG. 29 illustrates peak analysis of potential threats which show up in positive ionization.

While there is shown and described herein certain specific structures embodying various embodiments of the invention, it will be manifest to those skilled in the art that various modifications and rearrangements of the parts may be made without departing from the spirit and scope of the underlying inventive concept and that the same is not limited to the particular forms herein shown and described except insofar as indicated by the scope of the appended claims

What is claimed:

1. Apparatus for chemical trace detection comprising:
 - a spectrometry testing instrument comprising,
 - an inlet port configured to receive a vaporized analyte, and
 - a detector configured to analyze the vaporized analyte;
 - a swab having a distal end with an analyte release area, said analyte release area having an apex angle of 120 degrees or less; and
 - a swab holder comprising,
 - a housing configured to be mounted in front of the inlet port of said spectrometry testing instrument, said housing having a swab positioning slot configured to slidably receive said swab and position said swab

12

with said analyte release area adjacent to and directed toward the inlet; and

- a high voltage electrical source having
 - a ground in electrical communication with the inlet port; and
 - a delivery contact within said housing and having a terminal contact end engaging said swab when received within said swab positioning slot, whereby voltage is selectively applied to an analyte of a chemical sample on said swab and directs an ionized plume of said analyte into said inlet port, and wherein the high voltage electrical source is configured and arranged to apply said voltage such that there is a differential potential between said swab and said inlet port.

2. The apparatus of claim 1 wherein said apex angle is 45 degrees or less.

3. The apparatus of claim 1 wherein said spectrometry testing instrument is an IMS device or an MS device.

4. The apparatus of claim 1 wherein said terminal contact end engages said analyte release area when received within said swab positioning slot.

5. The apparatus of claim 1 further comprising a solvent.

6. The apparatus of claim 5 wherein said solvent includes an internal standard or an adducting agent.

7. The apparatus of claim 6 further comprising a solvent dispenser comprising a solvent supply container, a solvent dispenser nozzle positioned adjacent to said analyte release area and a pump in fluid communication with said solvent supply container and said solvent dispenser nozzle and selectively actuable to dispense a predetermined volume of solvent from said solvent supply container through said dispensing nozzle and onto said analyte release area.

8. The apparatus of claim 7 wherein said solvent dispenser is mounted to said housing of said swab holder.

9. The apparatus of claim 6 wherein said spectrometry testing instrument is an IMS device or an MS device.

10. The apparatus of claim 1 wherein said swab has a sampling end including a central preformed triangular analyte release area and flanking side portions.

11. The apparatus of claim 10, wherein said housing includes deflector arms which deflect the flanking side portions of said sampling end of said swab away from the analyte release area.

12. Apparatus for chemical trace detection comprising:

- a swab having a distal end with an analyte release area, said analyte release area having an apex angle of 120 degrees or less; and

a swab holder comprising,

- a housing including a swab positioning slot configured to slidably receive said swab and position said swab with said analyte release area adjacent to and directed toward an inlet port of a spectrometry testing instrument; and

a high voltage electrical source having

- a ground in electrical communication with the inlet port; and

a voltage delivery contact within said housing and having a terminal contact end engaging said swab when received within said swab positioning slot,

whereby voltage is selectively applied to an analyte of a chemical sample on said swab and directs an ionized plume of said analyte into said inlet port, and wherein the high voltage electrical source is configured and arranged to apply said voltage such that there is a differential potential between said swab and said inlet port.

13. The apparatus of claim 12 wherein said apex angle is 45 degrees or less.

14. The apparatus of claim 12 wherein said spectrometry testing instrument is an IMS device or an MS device.

15. The apparatus of claim 12 wherein said terminal contact end engages said analyte release area when received within said swab positioning slot. 5

16. The apparatus of claim 12 further comprising a solvent.

17. The apparatus of claim 16 wherein said solvent includes an internal standard or adducting agent. 10

18. The apparatus of claim 17 further comprising a solvent dispenser comprising a solvent supply container, a solvent dispenser nozzle positioned adjacent to said analyte release area and a pump in fluid communication with said solvent supply container and said solvent dispenser nozzle and selectively actuable to dispense a predetermined volume of solvent from said solvent supply container through said dispensing nozzle and onto said analyte release area. 15

19. The apparatus of claim 18 wherein said solvent dispenser is mounted to said housing of said swab holder. 20

20. The apparatus of claim 17 wherein said spectrometry testing instrument is an IMS device or an MS device.

21. The apparatus of claim 12 wherein said swab has a sampling end including a central preformed triangular analyte release area and flanking side portions. 25

22. The apparatus of claim 21, wherein said housing includes deflector arms which deflect the flanking side portions of said sampling end of said swab away from the analyte release area when said swab is received within said slot. 30

* * * * *